Message Passing-Based Channel Estimation and Symbol Detection for Clipped OCDM Systems

Mingyu Hu, Nan Wu, *Senior Member, IEEE*, Dongxuan He, *Member, IEEE*, Haoyang Li, and Tony Q. S. Quek, *Fellow, IEEE*

***Abstract*—Signal clipping effectively mitigates the high peak-to-average power ratio (PAPR) problem in multi-carrier systems like orthogonal chirp division multiplexing (OCDM). However, the nonlinear distortion induced by clipping severely degrades the performance of both channel estimation and symbol detection. To address this challenge, we propose a hybrid iterative receiver that combines clipping expectation propagation (CEP) and variance modification generalized approximate message passing (VMGAMP) for clipped OCDM systems. Specifically, by factorizing the *a posteriori* probability of the received signal, a unified factor graph is constructed, where the clipping factor serves as a gate block that schedules the message flow according to clipping state. Subsequently, a CEP-based channel estimator is developed, in which a clipping approach based on signal reconstruction is designed under the gate mechanism. To reduce computational complexity of moment matching, a truncated adaptive uniform-grid quadrature is employed. Furthermore, a VMGAMP-based detector is designed for symbol detection, alongside a refined VMGAMP (rVMGAMP) variant tailored for severe clipping scenarios. Variance modification is adopted to refine the variance estimation associated with the clipping distortion. Additionally, a Cramér-Rao lower bound (CRLB) is derived to provide a theoretical benchmark for the CEP-based estimator. Numerical results demonstrate that the proposed CEP-VMGAMP-based iterative receiver significantly outperforms baseline schemes for clipped OCDM systems.**

***Index Terms*—Clipping distortion, clipping expectation propagation, factor graph, iterative receiver, orthogonal chirp division multiplexing, variance modification generalized approximate message passing.**

TABLE I
LIST OF VARIABLE NOTATIONS

$\mathbf{b}$, $\mathbf{c}$, c_m^q, $\mathbf{c}_{m_c}$	Information-bit vector, coded-bit vector, q-th bit of x_m, and binary label of $\mathcal{M}_{m_c}$.
$\mathbf{x}$, x_m, $\hat{\mathbf{x}}$	Fresnel-domain symbol vector, its m-th entry, and its soft estimate.
$\mathbf{u}$, u_n, $\bar{\mathbf{u}}$, $\bar{u}_n$	Unclipped/clipped time-domain OCDM signals and their n-th entries.
$\mathbf{s}$, $\mathbf{r}$, $\mathbf{y}$, $\mathbf{y}_p$	Transmit signal, received signal, data observation, and pilot observation.
$\mathbf{w}$, $\mathbf{w}_p$, $\mathbf{w}^{\text{clip}}$	Data noise, pilot noise, and equivalent clipping distortion.
$\mathbf{x}_p$, $\tilde{\mathbf{x}}_p$	Fresnel-domain pilot and its time-domain counterpart.
$\mathbf{z}$, $\bar{\mathbf{z}}$	Channel-output variables before and after clipping.
$\mathbf{x}^{\text{ext}}$, $\mathbf{x}^{\text{like}}$, $\mathbf{x}^{\text{post}}$	Extrinsic, likelihood, and posterior symbol messages.
$\mathbf{u}^{\text{prior}}$, $\mathbf{u}^{\text{like}}$, $\mathbf{u}^{\text{post}}$	Prior, likelihood, and posterior time-domain messages.
$\mathbf{z}^{\text{prior}}$, $\mathbf{z}^{\text{like}}$, $\mathbf{z}^{\text{post}}$	Prior, likelihood, and posterior channel-output messages.
$\mathbf{h}_{\text{TD}}^{\text{prior}}$, $\mathbf{h}_{\text{TD}}^{\text{post}}$, $\mathbf{h}_{\text{TD},b}^{\text{like}}$	Time-domain channel prior, posterior, and feedback likelihood messages.
$\mathbf{h}_{\text{FD}}^{\text{prior}}$, $\mathbf{h}_{\text{FD}}^{\text{like}}$, $\mathbf{h}_{\text{FD}}^{\text{post}}$	Frequency-domain channel prior, likelihood, and posterior messages.
$m_{h_l}^a$, $v_{h_l}^a$, $m_{h_l}^t$, $v_{h_l}^t$	Prior and posterior mean/variance of channel tap h_l.
$m_{h_l\to f_n}^t$, $v_{h_l\to f_n}^t$	Mean/variance of the feedback message from h_l to f_n.
$\hat{b}_{f_n\to h_l}^t$, $\hat{\tau}_{f_n\to h_l}^t$	Mean/variance of the likelihood message from f_n to h_l.
$\hat{\tau}_{\text{clip},n}$, $\hat{\tau}_{\text{eff},n}$	Clipping-distortion variance and effective variance.
p_n^{var}, φ_n, $n_\parallel$, $n_\perp$	ICG pseudo-covariance, posterior phase, and radial/tangential distortion components.
$\hat{p}_n^t$, $\tau_n^{p,t}$, $\check{p}_n^t$, $\tau_n^{\check{p},t}$	Coupled and decoupled likelihood-message means/variances.
$\bar{u}_n^t$, $\tau_n^{\bar{u},t}$, $u_n^{\text{post},t}$, $\tau_n^{u,\text{post},t}$	Decoupled prior mean/variance and posterior mean/variance of u_n.
$\tilde{z}_n^t$, $\tau_n^{\tilde{z},t}$, s_n^t, $\tau_n^{s,t}$	Recoupled posterior mean/variance and GAMP residual mean/variance.
$\hat{r}_{x_m}^t$, $\tau_{x_m}^{r,t}$	Mean/variance of the likelihood message for symbol x_m.
$L_{\text{dec}}^{e,t}(c_m^q)$, $L_{\text{SD}}^{e,t}(c_m^q)$	Extrinsic log-likelihood ratios (LLRs) from the LDPC decoder and symbol detector.

I. INTRODUCTION

ORTHOGONAL chirp division multiplexing (OCDM) has been proposed as a novel multi-carrier transmission scheme for high-speed communication [1]. Unlike orthogonal frequency division multiplexing (OFDM), where each data symbol is carried by an individual subcarrier, the inverse discrete Fresnel transform (IDFnT) spreads each Fresnel-domain symbol over the entire OCDM block. This double-spreading property enables OCDM to exploit multipath diversity in frequency-selective channels, which improves its robustness to time-burst and narrowband interference. Moreover, the discrete Fresnel transform (DFnT)/IDFnT can be implemented using fast Fourier transform (FFT)/inverse FFT (IFFT) operations and diagonal phase rotations, allowing most of the conventional OFDM transceiver architecture to be retained with limited modification [2], [3], [4], and thereby making OCDM a potential candidate for next-generation mobile communications [5], [6]. Considering the aforementioned advantages, there is a wealth of literature on OCDM transmission in the fields of terrestrial wireless communication [7], integrated

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sensing and communication [8], underwater acoustic communication [9], [10], and optical communication [11], etc.

However, the implementation of OCDM is still challenging. In particular, many studies have analytically shown that OCDM experiences the same high peak-to-average power ratio (PAPR) problem as OFDM [4], leading to low energy efficiency and nonlinear distortion in the high power amplifier (HPA) [12], [13]. This issue becomes even more severe when the HPA is driven close to saturation in order to improve energy efficiency. To characterize this effect, substantial research efforts have modeled HPA nonlinearity using memoryless models [14] and Volterra-series models [15], [16], [17].

On the other hand, various PAPR reduction solutions have been proposed for multi-carrier systems [18], [19], [20], [21], [22]. With these techniques, the system can operate at a higher average power while maintaining an almost linear amplification property, thereby improving the energy efficiency of the HPA. For instance, the authors in [18] proposed a probabilistic technique along with predistortion, reducing the PAPR of OCDM in a joint radar-communication system. Additionally, the chirp-reservation-based [19] and tone-reservation-based [20] PAPR reduction schemes were presented, which efficiently reduce the PAPR of OCDM. Nevertheless, the above methods require reserving subcarriers, which deteriorates the spectral efficiency significantly. Accordingly, selected mapping (SLM) and partial transmit sequences (PTS) [21] are two classical methods for OFDM PAPR reduction without reserving dedicated subcarriers. However, these schemes require side information at the receiver, which introduces overhead and reduces the effective spectral efficiency. Alternatively, signal clipping is another effective approach to reduce PAPR by introducing clipping distortion into the transmission signal. Although this approach does not sacrifice spectral efficiency, it introduces both in-band and out-of-band interference, thereby degrading system performance [12].

To this end, numerous works have been dedicated to handling clipped multi-carrier systems. For example, an iterative compensation method was proposed to handle clipping distortion in clipped OFDM systems by suppressing the correlation among symbols. To further improve the detection performance, message passing-based receivers for clipped multi-carrier systems were proposed [23], [24]. In [23], a generalized approximate message passing (GAMP)-based receiver for nonlinearly distorted OFDM signals was proposed, which exhibits robust performance in frequency-selective fading channels. Further, a fast-GAMP (FGAMP)-based symbol detection method was proposed in [24], which mitigates clipping distortion through an iterative procedure with much lower complexity compared to the original GAMP and sub-optimal maximum likelihood (ML) algorithms. It is worth noting that the above GAMP/FGAMP receivers mainly address symbol detection for nonlinearly distorted OFDM signals. In particular, the receiver in [23] assumes perfect channel estimation and performs frequency-domain channel equalization separately before GAMP detection, whereas [24] primarily focuses on reducing the computational complexity of the GAMP detector. However, both methods are developed for nonlinearly distorted OFDM systems and do not incorporate the unknown channel coefficients and transmitted symbols into a unified iterative inference framework.

In the considered clipped OCDM system, the IDFnT spreads each Fresnel-domain symbol over the entire time-domain block, so time-domain clipping nonlinearly couples the Fresnel-domain symbols, while the resulting distortion is further mixed by the unknown frequency-selective channel. This coupling leads to a bidirectional dependence in which channel uncertainty affects clipping mitigation and symbol detection, while the reliability of data-aided channel estimation depends on the current soft symbol beliefs. Such a dependence motivates iterative soft-information exchange between the channel estimator and the symbol detector. To the best of our knowledge, joint channel estimation and symbol detection for OCDM in the presence of clipping distortion have not been explicitly addressed, which motivates the proposed receiver based on clipping expectation propagation (CEP) and variance modification generalized approximate message passing (VMGAMP).

Motivated by the above discussion, we propose a joint channel estimation and symbol detection receiver for clipped OCDM systems. Specifically, the proposed receiver consists of a CEP-based channel estimator, a VMGAMP-based symbol detector, and a refined VMGAMP (rVMGAMP) variant for severe clipping scenarios. The resulting iterative receivers are referred to as CEP-VMGAMP and CEP-rVMGAMP, respectively. The main contributions of this paper are summarized as follows:

- Based on the probabilistic model of OCDM signal, a message passing-based iterative receiver is developed for joint channel estimation and symbol detection. Unlike existing GAMP/FGAMP-based clipped-OFDM receivers that perform symbol detection with known or separately handled channel information, the proposed factor graph treats both the channel vector and the symbol vector as unknown variables and supports bidirectional soft-information exchange between channel estimation and symbol detection. We construct the factor graph by factorizing the *a posteriori* probability of the transmitted symbols and channel coefficients, where a clipping node with gate mechanism is used to control the message flow within the graph. Based on the factor graph, we design a cross-domain iterative receiver for clipped OCDM, which performs channel estimation, clipping distortion mitigation, and symbol detection in the frequency-domain, time-domain, and Fresnel-domain, respectively.
- Based on the properties of clipping, a CEP channel estimation algorithm is developed accordingly. To mitigate the impact of clipping distortion on channel estimation, a clipping approach based on signal reconstruction is utilized. Moreover, since the frequency-domain spreading of the OCDM signal makes moment matching computationally prohibitive for channel estimation, a truncated adaptive uniform-grid quadrature is developed to reduce computational complexity while maintaining satisfactory estimation accuracy.

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- To improve the reliability of symbol detection, we develop a VMGAMP-based symbol detector together with an rVMGAMP variant. To facilitate the distortion mitigation, a channel decoupling-coupling transformation is introduced to decouple the influence of channel spreading on clipped signal. In addition, clipping distortion mitigation method based on clipping detection and variance modification is embedded into the channel decoupling-coupling transformation. For the proposed VMGAMP framework, the circularly symmetric complex Gaussian (CSCG) model for effective distortion noise offers lower computational complexity, whereas the improper complex Gaussian (ICG) model provides better performance under severe clipping at the cost of a slight increase in computational complexity.
- To evaluate the performance of the proposed channel estimator, we derive a Cramér-Rao lower bound (CRLB) for channel estimation in clipped OCDM systems. The probability density function of the time-domain envelope of the multi-carrier signal is utilized to derive a closed-form expression for the power reduction, which is then applied to derive the CRLB. Furthermore, numerical results are provided to validate the effectiveness of the proposed joint channel estimation and symbol detection algorithms.

The rest of this paper is organized as follows. The system model is given in Section II. In Section III, the CEP channel estimation algorithm and the VMGAMP symbol detection algorithm are developed for clipped OCDM systems, and the proposed CEP-VMGAMP-based joint channel estimation and symbol detection algorithm is derived. In Section IV, the CRLB of the proposed channel estimation algorithm is derived. The performance of the proposed algorithms is evaluated in Section V. Finally, our conclusions are drawn in Section VI.

Notations: Boldface capital and lowercase letters denote matrices and vectors, respectively. Specifically, x_n and $\mathbf{X}_{m,n}$ denote the n-th element of the vector $\mathbf{x}$ and the (m,n)-th element of the matrix $\mathbf{X}$, respectively. The operations $(\cdot)^*$, $(\cdot)^T$, $(\cdot)^H$, $(\cdot)^{-1}$, and $\mathrm{tr}(\cdot)$ denote the complex conjugate, transpose, conjugate transpose, matrix inverse, and trace operation, respectively. I_L is the identity matrix with size $L \times L$, and $\mathcal{D}(\mathbf{x})$ represents a diagonal matrix constructed from the vector $\mathbf{x}$. The operator $\oslash$ is element-wise division, and $\Re\{x\}$ denotes the real part of x. $\mathcal{CN}(x; m_x, v_x)$ denotes a CSCG distribution of variable x with mean m_x and variance v_x. $\mathcal{CN}_{\mathrm{I}}(x; m_x, v_x, p_x)$ denotes an ICG distribution of variable x with mean m_x, variance v_x and pseudo-covariance p_x. The operator $\propto$ denotes equality up to a constant normalization factor. Furthermore, $\mathbb{E}\{\cdot\}$ denotes the expectation operation. $\tanh(\cdot)$ denotes hyperbolic tangent function. $p(\cdot)$ denotes a probability density function (PDF). Moreover, $\mathbb{Z}$ and $\mathbb{N}^+$ denote the set of integers and the set of positive integers, respectively. $\mathrm{clip}(\cdot)$ denotes the clipping operation. $\mathbf{1}_N$ and $\mathbf{0}_N$ denote the all-one column-vector and the all-zero column-vector of length N, respectively. The superscripts prior, like, post, and ext denote prior, likelihood, posterior, and extrinsic messages, respectively, while the subscripts TD, FD and SD denote the time-domain, the frequency-domain, and the symbol detection, respectively.

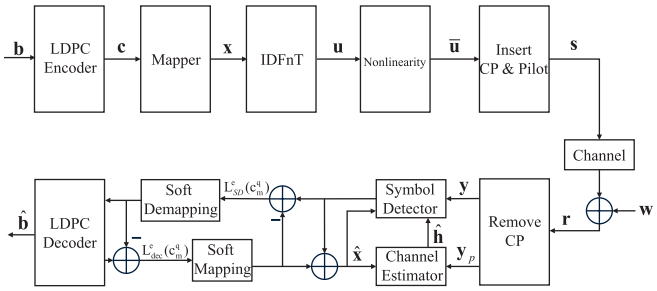


Fig. 1. The transceiver of the LDPC-coded OCDM system.

II. System Model

A. Transmitter

The transceiver of the low-density parity-check (LDPC)-coded OCDM system is shown in Fig. 1. At the transmitter, N_b information bits $\mathbf{b}$ are encoded, and N_c coded bits $\mathbf{c}$ are divided into N_{blk} groups, where each group contains $N_{\mathrm{bs}} = N_c/N_{\mathrm{blk}}$ bits, and block length is $N_d = N_{\mathrm{bs}}/Q$, where $Q = \log_2|\mathcal{M}|$. The coded bits are mapped to quadrature amplitude modulation (QAM) symbols, where the constellation set is $\mathcal{M} \triangleq \{\mathcal{M}_1, \ldots, \mathcal{M}_{M_c}\}$. Let $q \in \{1, \ldots, Q\}$ denote the bit index within one symbol. Here, each element $\mathcal{M}_{m_c}$ corresponds to a binary vector $\mathbf{c}_{m_c} = \left[c^1_{m_c}, \ldots, c^Q_{m_c}\right]^T$. Then, the IDFnT is used to transform the QAM symbols into OCDM symbol block $\mathbf{u}$, given by

$$\mathbf{u} = \mathbf{\Phi}^H \mathbf{x}, \tag{1}$$

where $\mathbf{x} \in \mathbb{C}^{N_d \times 1}$ contains the QAM symbols. Here, we assume the number of subcarriers in an OCDM symbol block is N_d. $\mathbf{\Phi}$ is the DFnT matrix with size $N_d \times N_d$, given by

$$\mathbf{\Phi}_{m,n} = \frac{1}{\sqrt{N_d}} e^{-j\frac{\pi}{4}} \times \begin{cases} e^{j\frac{\pi}{N_d}(m-n)^2} & N_d \equiv 0 (\mathrm{mod} 2) \\ e^{j\frac{\pi}{N_d}\left(m+\frac{1}{2}-n\right)^2} & N_d \equiv 1 (\mathrm{mod} 2). \end{cases} \tag{2}$$

Then, the signal is clipped according to the clipping threshold C_r, which can be expressed as

$$\bar{u}_n = \mathrm{clip}(u_n) = \begin{cases} u_n, & |u_n| \le C_r \\ C_r e^{j2\pi \angle u_n}, & |u_n| > C_r, \end{cases} \tag{3}$$

where u_n is the n-th element of $\mathbf{u}$ and $\angle u_n$ is the phase angle of u_n, and $n \in \{1, \ldots, N_d\}$. The clipping ratio is defined as C_r/σ [24], where σ^2 denotes the average power of each OCDM symbol and is normalized to unity.

Finally, a pilot and a cyclic prefix (CP) are inserted into each symbol block for channel estimation and mitigating the inter-symbol interference (ISI). Since the pilot is designed as a constant-envelope chirp sequence with amplitude below the clipping threshold, it is assumed to remain unclipped throughout this paper [10]. Specifically, $\widetilde{\mathbf{x}}_p = \mathbf{\Phi}_p^H \mathbf{x}_p$, where $\mathbf{\Phi}_p^H$ is the IDFnT matrix with size $N_p \times N_p$, and $\mathbf{x}_p = [\sqrt{N_p}, 0, \ldots, 0]^T$ with N_p being the length of the pilot.

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B. Receiver

At the receiver, the CP is removed. Accordingly, the signal $\mathbf{y}$ and pilot $\mathbf{y}_p$ are separated. The received data signal can be represented as

$$\begin{aligned}\mathbf{y} &= \mathbf{H}\bar{\mathbf{u}} + \mathbf{w} \\ &= \mathbf{H}\text{clip}(\mathbf{\Phi}^H\mathbf{x}) + \mathbf{w} \\ &= \mathbf{H}\mathbf{\Phi}^H\mathbf{x} + \mathbf{H}\mathbf{w}^{\text{clip}} + \mathbf{w},\end{aligned} \tag{4}$$

where $\mathbf{w}^{\text{clip}} \triangleq \text{clip}\left(\mathbf{\Phi}^H\mathbf{x}\right) - \mathbf{\Phi}^H\mathbf{x}$ denotes the clipping-distortion vector, and $\mathbf{H}$ is a column-wise circulant matrix of size $N_d \times N_d$. Although $\mathbf{w}^{\text{clip}}$ appears as an additive term in the above model, it is deterministically generated by $\mathbf{x}$ and is therefore not independent noise. The frequency-selective channel further mixes the clipping distortion across the time-domain symbols. Consequently, joint inference of the channel vector $\mathbf{h}$ and the symbol vector $\mathbf{x}$ is nonlinear and nonseparable, which motivates iterative soft-information exchange between channel estimation and symbol detection.

The first column of $\mathbf{H}$ is $[h_1, h_2, \ldots, h_L, 0, \ldots, 0]^T$, where h_l denotes the l-th channel impulse response (CIR) coefficient, $l \in \{1, \ldots, L\}$ is the channel-tap index, and L denotes the pre-specified maximum CIR length. Moreover, $\mathbf{w}$ denotes the additive white Gaussian noise (AWGN) vector. In addition, $\mathbf{y}_p = \mathbf{H}_p\mathbf{\Phi}_p^H\mathbf{x}_p + \mathbf{w}_p$ is the received pilot, where $\mathbf{H}_p$ is a column-wise circulant matrix of size $N_p \times N_p$.

III. The Proposed CEP-VMGAMP-Based Joint Channel Estimation and Symbol Detection Algorithm

In this section, we propose a CEP-VMGAMP-based joint channel estimation and symbol detection algorithm for clipped OCDM systems. The main idea of the proposed receiver is to divide the joint inference problem into three coupled functions. The clipping gate models the piecewise clipping relation and schedules messages according to the clipping state. The CEP estimator refines the pilot-initialized channel estimate by exploiting reconstructed soft symbols, while the VMGAMP/rVMGAMP detector performs channel decoupling-coupling, clipping-distortion mitigation, and soft symbol detection. These modules exchange channel and symbol messages in the turbo schedule summarized in Algorithm 3, together with decoder extrinsic information in LDPC-coded systems.

According to (4), the *a posteriori* probability of the channel parameters and transmitted symbols can be factorized as

$$\begin{aligned}&p(\mathbf{x}, \mathbf{h} \mid \mathbf{y}, \mathbf{y}_p) \\ &\propto p(\mathbf{y} \mid \bar{\mathbf{z}})p(\bar{\mathbf{z}} \mid \mathbf{h}, \bar{\mathbf{u}})p(\bar{\mathbf{u}} \mid \mathbf{u})p(\mathbf{u} \mid \mathbf{x})p(\mathbf{h})p(\mathbf{x}) \\ &\propto \prod_n p(y_n \mid \bar{z}_n)p(\bar{z}_n \mid \mathbf{h}, \bar{\mathbf{u}}) \prod_{n_u} p(\bar{u}_{n_u} \mid u_{n_u})p(u_{n_u} \mid \mathbf{x}) \\ &\quad \times \prod_m p(x_m)p(\mathbf{h}|\mathbf{x}_p, \mathbf{y}_p),\end{aligned} \tag{5}$$

where $p(y_n \mid \bar{z}_n) \sim \mathcal{CN}(y_n; \bar{z}_n, \sigma_n^2)$ is the likelihood function, $p(\bar{\mathbf{z}} \mid \mathbf{h}, \bar{\mathbf{u}})$ is a delta function defined by linear transform $\bar{\mathbf{z}} = \mathbf{H}\bar{\mathbf{u}}$, $p(\bar{\mathbf{u}} \mid \mathbf{u})$ delta function is defined by clipping in (3), $p(\mathbf{u} \mid \mathbf{x})$ is defined by OCDM modulation in (1), $p(\mathbf{h})$ is the *a priori* probability of the channel parameters, $p(\mathbf{x})$ is the *a priori* probability of transmitted symbols, and σ_n^2 is the variance of noise. $p(\mathbf{h})$ denotes the pilot-initialized Gaussian prior approximation of the channel parameters, which is approximated as $p(\mathbf{h}) \approx p(\mathbf{h}|\mathbf{x}_p, \mathbf{y}_p)$. It is constructed from the pilot observation model and will be specified in Section III-B.

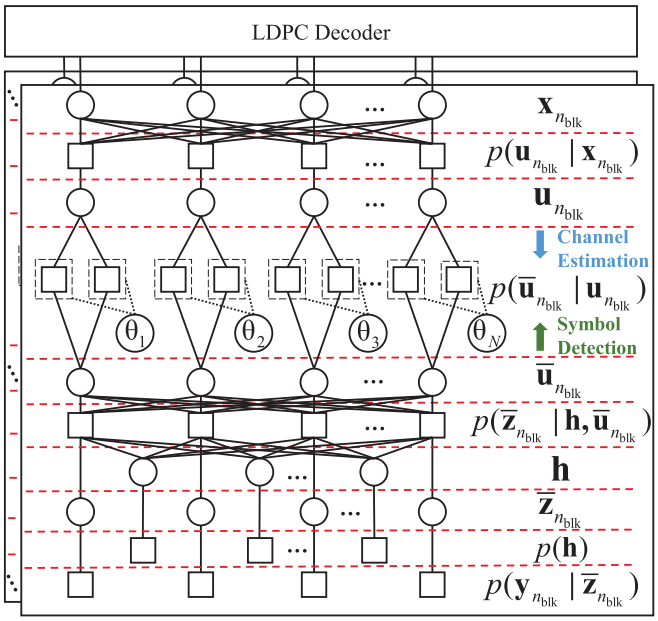


Fig. 2. Factor graph for joint channel estimation and symbol detection of OCDM signal with clipping distortion.

A. Message Passing for Signal Clipping

Based on the factorization of the *a posteriori* probability in (5), a factor graph is constructed. As shown in Fig. 2, the factor graph characterizes the statistical relationships between variables, where the squares denote the factor nodes (FN) and the circles denote the variable nodes (VN). Additionally, each frame consists of N_{blk} OCDM symbol blocks, and each symbol block corresponds to a slice shown in Fig. 2. The clipping nodes in Fig. 2 follow the standard gate notation of [25], where $\theta_n = 0$ and $\theta_n = 1$ activate the unclipped and clipped branches, respectively. The belief propagation (BP) update equations and the corresponding messages at the clipping factor are provided in Appendix A.

As $p(\bar{u}_n \mid u_n)$ represents a piecewise function, we introduce a gate mechanism in [25] into the clipping node to control the flow of messages. Specifically, θ_n is a selector variable indicating which branch of the clipping function is active. Therefore, $p(\bar{u}_n \mid u_n, \theta_n)$ can be represented as

$$p(\bar{u}_n \mid u_n, \theta_n) = \delta(\bar{u}_n - C_r e^{j2\pi\angle u_n})^{\theta_n}\delta(\bar{u}_n - u_n)^{1-\theta_n}, \tag{6}$$

where $\theta_n = \mathbb{I}(|u_n| > C_r)$ is a binary latent random variable induced by the random variable u_n, and $\delta(\cdot)$ denotes the Dirac delta function. The mapping from u_n to θ_n is deterministic, but θ_n remains random because u_n is random and unknown at the receiver. $\mathbb{I}$ is the indicator function, which equals 1 if the event is true and 0 otherwise.

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$p(\bar{u}_n|u_n) = \sum_{\theta_n\in\{0,1\}} p(\theta_n|u_n)p(\bar{u}_n|u_n,\theta_n)$ is equivalent to the original piecewise clipping function in (3). Following the standard gate representation in [25], the realization of θ_n determines which clipping branch is active. When the signal is not clipped, i.e., $\theta_n = 0$, the FN $p(\bar{u}_n \mid u_n)$ is actually an equality constraint and the messages can be computed on the factor graph in the same manner as in conventional message passing. However, once the selector detects that the observation $\bar{\mathbf{u}}$ is clipped, the gates adaptively route the messages propagating between $\bar{\mathbf{u}}$ and $\mathbf{u}$ along the clipping piece, and the subsequent message is computed by considering the clipping property. Considering the effect of clipping distortion on the signal, we model the nonlinear segment of the piecewise function by $\mathbf{y} = \mathbf{z} + \mathbf{H}\mathbf{w}^{\text{clip}} + \mathbf{w}$, where $\mathbf{w}^{\text{clip}}$ represents the distortion induced by clipping.

The key issue is that, if the clipping-induced uncertainty is ignored, a clipped sample is processed using only the AWGN variance σ_n^2. This underestimates the effective uncertainty and yields an over-confident likelihood message. To address this challenge, the effective error is approximated as a CSCG variable, i.e., $\mathbf{H}\mathbf{w}^{\text{clip}} + \mathbf{w} \sim \mathcal{CN}(\mathbf{H}\mathbf{w}^{\text{clip}} + \mathbf{w}; \mathbf{0}_{N_d}, \mathbf{1}_{N_d} \oslash \boldsymbol{\epsilon}^y)$, and the precision $\boldsymbol{\epsilon}^y$ is introduced as a hidden random variable, and $\mathbf{y} \sim \mathcal{CN}(\mathbf{y}; \mathbf{z}, \mathbf{1} \oslash \boldsymbol{\epsilon}^y)$. According to the mean-field (MF) rules, the message passed from factor node f_{y_n} to variable node ϵ_n^y is given by

$$\begin{aligned} m_{f_{y_n}\to\epsilon_n^y}(\epsilon_n^y) &\propto \exp\left\{\mathbb{E}_{z_n}[\log f_{y_n}(y_n \mid z_n, \epsilon_n^y)]\right\} \\ &\propto \epsilon_n^y \exp\left\{-\epsilon_n^y\left(|y_n - \hat{p}_n|^2 + \tau_n^{\hat{p}}\right)\right\}, \qquad (7) \\ b(\epsilon_n^y) &\propto f_{\epsilon_n^y}(\epsilon_n^y) m_{f_{y_n}\to\epsilon_n^y}(\epsilon_n^y), \qquad (8) \end{aligned}$$

where $f_{\epsilon_n^y}(\epsilon_n^y) \propto 1/\epsilon_n^y$, and the mean of noise precision ϵ_n^y is

$$\begin{aligned} \hat{\epsilon}_n^y &= \int_0^\infty \epsilon_n^y b(\epsilon_n^y) d\epsilon_n^y \\ &= \frac{1}{|y_n - \hat{p}_n|^2 + \tau_n^{\hat{p}}}. \qquad (9) \end{aligned}$$

Since the Gaussian message of z_n is characterized by mean $\hat{p}_n$ and variance $\tau_n^{\hat{p}}$, the denominator in (9) is the residual second moment, i.e.,

$$\mathbb{E}_{z_n}\left[|y_n - z_n|^2\right] = |y_n - \hat{p}_n|^2 + \tau_n^{\hat{p}}. \qquad (10)$$

Therefore, $(\hat{\epsilon}_n^y)^{-1}$ estimates the aggregate power of the clipping distortion and AWGN. Since the AWGN variance σ_n^2 is known, the excess residual power is attributed to clipping distortion.

The received signal is corrupted by both additive noise and clipping distortion, and the variance of clipping distortion is expressed as

$$\hat{\tau}_{\text{clip},n} = \frac{\text{tr}((\mathbf{H}^{-1})^H\mathbf{H}^{-1})}{N_d} \max\left(0, |y_n - \hat{p}_n|^2 + \tau_n^{\hat{p}} - \sigma_n^2\right), \qquad (11)$$

$$\hat{\tau}_{\text{eff},n} = \hat{\tau}_{\text{clip},n} + \tau_n^{\bar{u}}, \qquad (12)$$

where $\bar{u}_n$ and $\tau_n^{\bar{u}}$ are derived later. It should be emphasized that the above CSCG model is a tractable second-order approximation rather than the exact distribution of the signal-dependent clipping distortion. Higher-order non-Gaussian statistics are not explicitly represented by this approximation.

The CSCG approximation is suitable when clipping is mild or the effective error is dominated by AWGN, for which the radial and tangential uncertainties are comparable. Under severe clipping and high signal-to-noise ratio (SNR) conditions, signal clipping mainly introduces uncertainty along the radial direction, resulting in nonzero pseudo-covariance. In this regime, the CSCG approximation causes a noticeable performance loss. Therefore, the ICG model is adopted to preserve this second-order noncircularity. Based on the above derivation for CSCG, the corresponding message updates under the ICG model are further derived [27]. Considering the CSCG and ICG distributions, the proper messages and improper messages are expressed as

$$u_n \sim \mathcal{CN}(u_n; \breve{p}_n^t, \tau_n^{\breve{p},t}), \qquad (13)$$

$$\begin{aligned} \bar{u}_n &\sim \mathcal{CN}_{\text{I}}(\bar{u}_n; \bar{u}_n^t, (\tau_n^{\bar{u},t} + \hat{\tau}_{\text{clip},n}^t), p_n^{\text{var},t}) \\ &\propto \exp\left(-(u_a - \mu_a)^H C^{-1}(u_a - \mu_a)\right), \qquad (14) \end{aligned}$$

where $\breve{p}_n$, $\tau_n^{\breve{p}}$, $\bar{u}_n^t$, $\tau_n^{\bar{u},t}$ are derived later. The derivation of the corresponding covariance model is provided in Appendix B. The resulting posterior variance, pseudo-covariance, and mean are given below. Under severe clipping, the ICG model retains the clipping-induced pseudo-covariance $p_n^{\text{var},t} = \hat{\tau}_{\text{clip},n}^t e^{j2\phi_n^t}$. ϕ_n^t is the *a posteriori* phase angle, which is expressed as

$$\phi_n^t = \frac{\frac{\tau_n^{\bar{u},t}}{2|\bar{u}_n^t|^2}\angle\breve{p}_n^t + \frac{\tau_n^{\breve{p},t}}{2|\breve{p}_n^t|^2}\angle\bar{u}_n^t}{\frac{\tau_n^{\bar{u},t}}{2|\bar{u}_n^t|^2} + \frac{\tau_n^{\breve{p},t}}{2|\breve{p}_n^t|^2}}, \qquad (15)$$

where $\frac{\tau_n^{\bar{u},t}}{2|\bar{u}_n^t|^2}$ and $\frac{\tau_n^{\breve{p},t}}{2|\breve{p}_n^t|^2}$ are the variances of $\angle\bar{u}_n^t$ and $\angle\breve{p}_n^t$, respectively. The determinant of covariance matrix is

$$\begin{aligned} D &\triangleq (\tau_n^{\bar{u},t} + \hat{\tau}_{\text{clip},n}^t)^2 - |\hat{\tau}_{\text{clip},n}^t e^{j2\phi_n^t}|^2 \qquad (16) \\ &= \tau_n^{\bar{u},t}(\tau_n^{\bar{u},t} + 2\hat{\tau}_{\text{clip},n}^t) > 0. \end{aligned}$$

The *a posteriori* covariance and pseudo-covariance are

$$v_{\text{post}} = \tau_n^{u^{\text{post},t}} = \frac{\alpha}{\Delta}, \qquad (17)$$

$$p_{\text{post}} = \frac{-\beta}{\Delta}, \qquad (18)$$

where $\alpha = \frac{1}{\tau_n^{\breve{p},t}} + \frac{\tau_n^{\bar{u},t} + \hat{\tau}_{\text{clip},n}^t}{D}$, $\beta = -\frac{\hat{\tau}_{\text{clip},n}^t e^{j2\phi_n^t}}{D}$, $\Delta = \alpha^2 - |\beta|^2$, and the *a posteriori* mean is

$$\mu_{\text{post}} = u_n^{\text{post},t} = v_{\text{post}}\breve{\eta} + p_{\text{post}}\breve{\eta}^*, \qquad (19)$$

where

$$\breve{\eta} = \frac{\breve{p}_n^t}{\tau_n^{\breve{p},t}} + \frac{(\tau_n^{\bar{u},t} + \hat{\tau}_{\text{clip},n}^t)\bar{u}_n^t - \hat{\tau}_{\text{clip},n}^t e^{j2\phi_n^t}(\bar{u}_n^t)^*}{\tau_n^{\bar{u},t}(\tau_n^{\bar{u},t} + 2\hat{\tau}_{\text{clip},n}^t)}. \qquad (20)$$

B. CEP Channel Estimation Algorithm

To estimate channel parameters and detect data symbols accurately, a CEP-VMGAMP-based joint channel estimation and symbol detection algorithm is proposed. More specifically, the pilot is used to obtain the least-squares (LS) estimate of the channel, which is then employed to construct a Gaussian prior approximation of the channel parameters. Then, the likelihood and the *a priori* information of the symbols are utilized for

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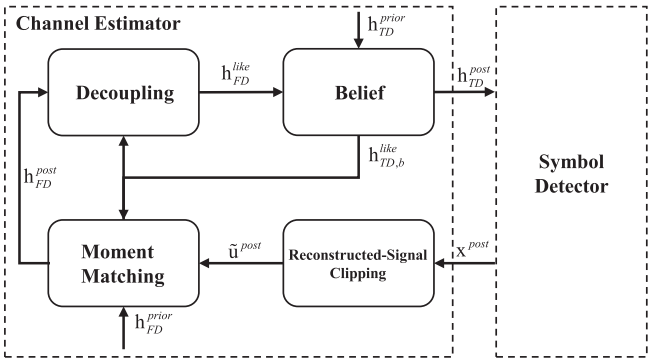


Fig. 3. The message flows of the CEP-based channel estimator.

data-aided channel estimation, which are given by symbol detector and LDPC decoder, respectively.

Fig. 3 illustrates the message flows of the channel estimation algorithm. The input messages consist of the *a posteriori* messages $\mathbf{x}^{\text{post}}$ (i.e., $\hat{\mathbf{x}}^{\text{post},t}$ and $\hat{\boldsymbol{\tau}}_{\mathbf{x}}^{\text{post},t}$) from the signal detector, the time-domain *a priori* messages $\mathbf{h}_{\text{TD}}^{\text{prior}}$ from the pilot-based channel estimation, and the frequency-domain *a priori* messages $\mathbf{h}_{\text{FD}}^{\text{prior}}$ derived from the input signal. The output messages are the *a posteriori* channel message $\mathbf{h}_{\text{TD}}^{\text{post}}$ passed to the symbol detector, the frequency-domain likelihood message $\mathbf{h}_{\text{FD}}^{\text{like}}$, and the feedback likelihood message $\mathbf{h}_{\text{TD},b}^{\text{like}}$ used in the next CEP update.

For ease of understanding, the CEP channel estimation algorithm is decomposed into four steps as follows:

1) **Reconstructed-signal clipping step** (Line 5 in **Algorithm** 1) computes the messages $\hat{m}_{\widetilde{u}_n}^t$ and $\hat{v}_{\widetilde{u}_n}^t$.
2) **Moment matching step** (Line 6 in **Algorithm** 1) computes the messages $\hat{m}_{\Xi_{n,l}h_l}^t$ and $\hat{v}_{\Xi_{n,l}h_l}^t$.
3) **Decoupling step** (Line 8 in **Algorithm** 1) computes the messages $\hat{b}_{f_n\to h_l}^t$ and $\hat{\tau}_{f_n\to h_l}^t$.
4) **Belief step** (Line 12, 14 in **Algorithm** 1) computes the messages $m_{h_l}^t$, $v_{h_l}^t$, $m_{h_l\to f_n}^t$ and $v_{h_l\to f_n}^t$.

From the pilot observation model, the LS estimate is first obtained and then used to construct a Gaussian prior approximation $p(\mathbf{h}) \approx \mathcal{CN}(\mathbf{h}; \mathbf{m}_{\mathbf{h}}^a, \mathbf{v}_{\mathbf{h}}^a)$, which is expressed as

$$\mathbf{m}_{\mathbf{h}}^a = \frac{1}{\sqrt{N_p}} \mathbf{\Phi}_p \mathbf{y}_p, \tag{21}$$

$$\mathbf{v}_{\mathbf{h}}^a = \frac{1}{N_p} \sigma_n^2 \mathbf{1}_{N_p}. \tag{22}$$

The detailed expectation propagation (EP) moment-matching derivation of $\hat{m}_{\Xi_{n,l}h_l}^t$, $\hat{v}_{\Xi_{n,l}h_l}^t$, and ρ_n^t is provided in Appendix C. These expectations are evaluated using the truncated adaptive uniform-grid quadrature described below.

Because of the time-frequency domain spreading property of OCDM systems, the computational complexity of marginalization on frequency-domain symbols is $\mathcal{O}(|\mathcal{M}|^{N_d})$, which results in prohibitively high computational complexity. Therefore, a truncated adaptive uniform-grid quadrature is proposed to handle this issue, where the sampling region is selected as a rectangular area centered at $\hat{m}_{\widetilde{u}_n}^t$ with side length $2\sqrt{\hat{v}_{\widetilde{u}_n}^t/2}$ along both the real and imaginary dimensions. The expectation is approximated as

$$\begin{aligned}\mathbb{E}_{\widetilde{u}_n}\left[f\left(\widetilde{u}_n\right)\right] &= \sum_{\widetilde{u}_n} f\left(\widetilde{u}_n\right) p_m(\widetilde{u}_n) \\ &\approx \int f\left(\widetilde{u}_n'\right) p(\widetilde{u}_n') d\widetilde{u}_n' \\ &\approx \sum_{\widetilde{u}_n''} f\left(\widetilde{u}_n''\right) p_m(\widetilde{u}_n''),\end{aligned} \tag{23}$$

where $\widetilde{u}_n$ is approximated as $\widetilde{u}_n'$, $\widetilde{u}_n''$ is obtained by uniform-grid sampling in the truncated integration region of $\widetilde{u}_n'$, and $p_m(\widetilde{u}_n'')$ is the probability mass function (PMF) of $\widetilde{u}_n''$. Note that the computational complexity of the proposed truncated adaptive uniform-grid quadrature is independent of the modulation order. With $\hat{m}_{\Xi_{n,l}h_l}^t$ and $\hat{v}_{\Xi_{n,l}h_l}^t$, the messages from factor node f_n to variable node h_l can be represented as

$$\mu_{f_n\to h_l}^t(h_l) = \mathcal{CN}\left(\Xi_{n,l}h_l; \hat{b}_{f_n\to h_l}^t, \hat{\tau}_{f_n\to h_l}^t\right), \tag{24}$$

where the likelihood messages $\mathbf{h}_{\text{FD}}^{\text{like}}$ (i.e., $\hat{b}_{f_n\to h_l}^t$ and $\hat{\tau}_{f_n\to h_l}^t$) can be represented as

$$\begin{aligned}\hat{b}_{f_n\to h_l}^t &= \left(\rho_n^t v_{h_l\to f_n}^{t-1}\right)^{-1}\left(\hat{m}_{\Xi_{n,l}h_l}^t - \frac{\hat{v}_{\Xi_{n,l}h_l}^t}{v_{h_l\to f_n}^{t-1}}\Xi_{n,l} m_{h_l\to f_n}^{t-1}\right) \\ &= \mathbb{E}_{\widetilde{u}_n}\left[\frac{\check{b}_{f_n}^t(\widetilde{u}_n)}{\tau_{f_n}^t(\widetilde{u}_n)}\right]\left(\rho_n^t\right)^{-1} + \Xi_{n,l} m_{h_l\to f_n}^{t-1},\end{aligned} \tag{25}$$

$$\begin{aligned}\hat{\tau}_{f_n\to h_l}^t &= \left((\hat{v}_{\Xi_{n,l}h_l}^t)^{-1} - (v_{h_l\to f_n}^{t-1})^{-1}\right)^{-1} \\ &= (\rho_n^t)^{-1} - v_{h_l\to f_n}^{t-1}.\end{aligned} \tag{26}$$

According to the BP rules, the *a posteriori* probability distribution of h_l is expressed as

$$\begin{aligned}\mu_{h_l}^t(h_l) &= p(h_l)\prod_n \mu_{f_n\to h_l}^t(h_l) \\ &= \mathcal{CN}\left(h_l; m_{h_l}^t, v_{h_l}^t\right),\end{aligned} \tag{27}$$

where the *a posteriori* messages $\mathbf{h}_{\text{TD}}^{\text{post}}$ (i.e., $m_{h_l}^t$ and $v_{h_l}^t$) can be represented as

$$m_{h_l}^t = v_{h_l}^t\left(\frac{m_{h_l}^a}{v_{h_l}^a} + \sum_n \Xi_{n,l}^* \frac{\hat{b}_{f_n\to h_l}^t}{\hat{\tau}_{f_n\to h_l}^t}\right), \tag{28}$$

$$v_{h_l}^t = \left((v_{h_l}^a)^{-1} + \sum_n (\hat{\tau}_{f_n\to h_l}^t)^{-1}\right)^{-1}, \tag{29}$$

where $m_{h_l}^a$ and $v_{h_l}^a$ denote the *a priori* messages. The messages from variable node h_l to factor node f_n can be represented as

$$\begin{aligned}\mu_{h_l\to f_n}^t(h_l) &= p(h_l)\prod_{n'\neq n} \mu_{f_{n'}\to h_l}^t(h_l) \\ &= \mathcal{CN}\left(h_l; m_{h_l\to f_n}^t, v_{h_l\to f_n}^t\right),\end{aligned} \tag{30}$$

where the feedback likelihood messages $\mathbf{h}_{\text{TD},b}^{\text{like}}$ (i.e., $m_{h_l\to f_n}^t$ and $v_{h_l\to f_n}^t$) can be represented as

$$m_{h_l\to f_n}^t = v_{h_l\to f_n}^t\left(\frac{m_{h_l}^t}{v_{h_l}^t} - \Xi_{n,l}^* \frac{\hat{b}_{f_n\to h_l}^t}{\hat{\tau}_{f_n\to h_l}^t}\right)$$

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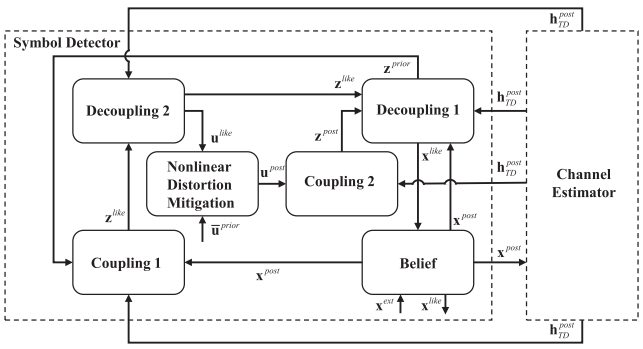


Fig. 4. The message flows of the VMGAMP-based symbol detector.

$$= m_{h_l}^t + \frac{v_{h_l}^t \left(m_{h_l}^t - \Xi_{n,l}^* \hat{b}_{f_n \to h_l}^t \right)}{\hat{\tau}_{f_n \to h_l}^t - v_{h_l}^t}, \tag{31}$$

$$v_{h_l \to f_n}^t = \left((v_{h_l}^a)^{-1} + \sum_{n' \neq n} (\hat{\tau}_{f_{n'} \to h_l}^t)^{-1} \right)^{-1} = \left((v_{h_l}^t)^{-1} - (\hat{\tau}_{f_n \to h_l}^t)^{-1} \right)^{-1}. \tag{32}$$

According to the aforementioned derivations, the CEP channel estimation algorithm for clipped OCDM signal is summarized in **Algorithm** 1.

C. VMGAMP/rVMGAMP Symbol Detection Algorithms

Ignoring the clipping distortion, the conventional GAMP algorithm fails to detect clipped OCDM signal accurately. Additionally, the time-domain spreading property of frequency-selective channels affects the characteristics of clipped OCDM signal, which imposes great challenges on clipping distortion mitigation. To tackle these issues, the VMGAMP and the rVMGAMP symbol detection algorithms are developed.

Fig. 4 illustrates the message flows of the VMGAMP symbol detection algorithm. Here, the input messages consist of the *a priori* messages $\mathbf{x}^{ext}$ (i.e., $L_{\text{dec}}^{e,t}(c_m^q)$) from LDPC decoder, the *a priori* messages $\bar{\mathbf{u}}^{\text{prior}}$ from received data signals, and the *a posteriori* messages $\mathbf{h}_{\text{TD}}^{\text{post}}$ from channel estimator. The detector outputs the symbol likelihood message $\mathbf{x}^{\text{like}}$ to the LDPC decoder and the *a posteriori* symbol message $\mathbf{x}^{\text{post}}$ to the CEP channel estimator and the next detector iteration.

For ease of understanding, the VMGAMP symbol detection algorithm is decomposed into six steps as follows:

1) **Belief step** computes the messages $\hat{x}_m^{\text{post},t}$ and $\hat{\tau}_{x_m}^{\text{post},t}$.
2) **Coupling 1 step** (Line 4 in **Algorithm** 2) computes the messages $\hat{p}_n^t$ and $\tau_n^{p,t}$.
3) **Decoupling 2 step** (Line 5, 6 in **Algorithm** 2) computes the messages $\check{\mathbf{p}}^t$, $\boldsymbol{\tau}^{\check{\mathbf{p}},t}$, $\bar{\mathbf{u}}^t$ and $\boldsymbol{\tau}^{\bar{\mathrm{u}},t}$.
4) **Nonlinear distortion mitigation step** (Line 7–16 in **Algorithm** 2) computes the messages $u_n^{\text{post},t}$ and $\tau_n^{u^{\text{post}},t}$.
5) **Coupling 2 step** (Line 17 in **Algorithm** 2) computes the messages $\widetilde{\mathbf{z}}^t$ and $\boldsymbol{\tau}^{\widetilde{z},t}$.
6) **Decoupling 1 step** (Line 18, 21 in **Algorithm** 2) computes the messages s_n^t, $\tau_n^{s,t}$, $\hat{r}_{x_m}^t$ and $\tau_{x_m}^{r,t}$.

According to the BP rules, the messages in symbol detection can be represented as

$$\begin{aligned} \mu_{z_n \to x_m}^t(x_m) &= \log(p(y_n \mid z_n)\mathcal{CN}(z_n; \hat{p}_{z_n \to x_m}^t, \tau_{z_n \to x_m}^{p,t})) \\ &= \log p(y_n \mid z_n) + \sum_{m' \neq m} \mu_{x_{m'} \to z_n}^t(z_n), \quad (33) \\ \mu_{x_m \to z_n}^{t+1}(x_m) &= \log(p(x_m \mid d_m)\mathcal{CN}(x_m; \hat{r}_{x_m \to z_n}^t, \tau_{x_m \to z_n}^{r,t})) \\ &= \log p(x_m \mid d_m) + \sum_{n' \neq n} \mu_{z_{n'} \to x_m}^t(x_m), \end{aligned} \tag{34}$$

where d_m denotes the *a priori* decoder information associated with x_m.

Based on the GAMP rules, $\mathbf{z}^{\text{like}}$ (i.e., $\hat{p}_n^t$ and $\tau_n^{p,t}$) is the likelihood messages, given by

$$\hat{p}_n^t = \sum_m A_{n,m} \hat{x}_m^{\text{post},t} - \tau_n^{p,t} \hat{s}_n^{t-1}, \tag{35}$$

$$\tau_n^{\hat{p},t} = \sum_m |A_{n,m}|^2 \hat{\tau}_{x_m}^{\text{post},t}, \tag{36}$$

where the transform matrix $\mathbf{A} = \mathbf{H}\boldsymbol{\Phi}^H$. To mitigate clipping distortion effectively, the time-domain OCDM signal is decoupled to avoid the effects of the channel. The likelihood messages $\mathbf{u}^{\text{like}}$ (i.e., $\check{\mathbf{p}}^t$ and $\boldsymbol{\tau}^{\check{\mathbf{p}},t}$) are represented as

$$\check{\mathbf{p}}^t = \mathbf{H}^{-1}\hat{\mathbf{p}}^t, \tag{37}$$

$$\boldsymbol{\tau}^{\check{p},t} = \frac{\text{tr}((\mathbf{H}^{-1})^H \mathbf{H}^{-1})}{N_d} \boldsymbol{\tau}^{\hat{p},t}, \tag{38}$$

and the *a priori* messages $\bar{\mathbf{u}}^{\text{prior}}$ (i.e., $\bar{\mathbf{u}}$ and $\boldsymbol{\tau}^{\bar{\mathrm{u}}}$) are represented as

$$\bar{\mathbf{u}}^t = \mathbf{H}^{-1}\mathbf{y}, \tag{39}$$

$$\tau_n^{\bar{\mathrm{u}},t} = \frac{\text{tr}((\mathbf{H}^{-1})^H \mathbf{H}^{-1})}{N_d} \sigma_n^2, \tag{40}$$

where $\mathbf{H}^{-1} = (\mathbf{H}^H\mathbf{H})^{-1}\mathbf{H}^H$ denotes the direct pseudo-inverse for a well-conditioned channel matrix. For an ill-conditioned channel matrix with a large condition number, the direct pseudo-inverse of $\mathbf{H}$ may significantly amplify the additive noise. Therefore, $\mathbf{H}^{-1}$ in (37)–(38) is replaced by

$$\mathbf{W}_{\text{LMMSE},\hat{p}} = \left(\mathbf{H}^H\mathbf{H} + \frac{1}{\sigma_u^2} \mathbf{I}_{N_d} \boldsymbol{\tau}^{\hat{p},t} \right)^{-1} \mathbf{H}^H, \tag{41}$$

and $\mathbf{H}^{-1}$ in (39)–(40) is replaced by

$$\mathbf{W}_{\text{LMMSE},\bar{u}} = \left(\mathbf{H}^H\mathbf{H} + \frac{\sigma_n^2}{\sigma_{\bar{u}}^2} \mathbf{I}_{N_d} \right)^{-1} \mathbf{H}^H, \tag{42}$$

where $\sigma_{\bar{u}}^2$ and σ_u^2 are the average power of the clipped OCDM signal and the average power of the unclipped OCDM signal, respectively.

The *a posteriori* messages $\mathbf{u}^{\text{post}}$ (i.e., $u_n^{\text{post},t}$ and $\tau_n^{u^{\text{post}},t}$) in VMGAMP are represented as

$$u_n^{\text{post},t} = \mathbb{E}\left\{ u_n \mid \bar{u}_n^t, \tau_n^{\bar{u},t}, \check{p}_n^t, \tau_n^{\check{p},t} \right\} = \frac{\tau_n^{\bar{u},t} \check{p}_n^t + \tau_n^{\check{p},t} \bar{u}_n^t}{\tau_n^{\check{p},t} + \tau_n^{\bar{u},t}}$$

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$$= \breve{p}_n^t + \frac{\tau_n^{\breve{p},t}}{\tau_n^{\breve{p},t} + \tau_n^{\widetilde{u},t}} \left(\bar{u}_n^t - \breve{p}_n^t\right), \tag{43}$$

$$\tau_n^{u^{\text{post}},t} = \mathbb{V}\left\{u_n \mid \bar{u}_n^t, \tau_n^{\bar{u},t}, \breve{p}_n^t, \tau_n^{\breve{p},t}\right\} = \frac{\tau_n^{\breve{p},t}\tau_n^{\widetilde{u},t}}{\tau_n^{\breve{p},t} + \tau_n^{\widetilde{u},t}}, \tag{44}$$

where

$$\tau_n^{\widetilde{u},t} = \begin{cases} \tau_n^{\bar{u},t} & |\breve{p}_n^t| \le C_r \\ \hat{\tau}_{\text{clip},n}^t + \tau_n^{\bar{u},t} & |\breve{p}_n^t| > C_r, \end{cases} \tag{45}$$

where $\hat{\tau}_{\text{clip},n}^t$ is defined in (11), and the *a posteriori* messages $\mathbf{u}^{\text{post}}$ in refined VMGAMP (rVMGAMP) are represented as (17) and (19) when $|\breve{p}_n^t| > C_r$. The variance modification has a direct reliability interpretation. Specifically, the posterior mean in (43) can be rewritten as $u_n^{\text{post},t} = \breve{p}_n^t + \omega_n^t\left(\bar{u}_n^t - \breve{p}_n^t\right), \omega_n^t = \frac{\tau_n^{\breve{p},t}}{\tau_n^{\breve{p},t}+\tau_n^{\widetilde{u},t}}$. For an unclipped sample, $\tau_n^{\widetilde{u},t} = \tau_n^{\bar{u},t}$. For a clipped sample, $\tau_n^{\widetilde{u},t} = \tau_n^{\bar{u},t} + \hat{\tau}_{\text{clip},n}^t$. Hence, a larger clipping-induced residual increases the effective observation variance and decreases ω_n^t, thereby reducing the weight assigned to the unreliable clipped amplitude. Thus, the proposed variance modification acts as a residual-based reliability correction.

The *a posteriori* messages $\mathbf{z}^{\text{post}}$ (i.e., $\widetilde{\mathbf{z}}^t$ and $\boldsymbol{\tau}^{\widetilde{z},t}$) are represented as

$$\widetilde{\mathbf{z}}^t = \mathbf{H}\mathbf{u}^{\text{post},t}, \tag{46}$$

$$\boldsymbol{\tau}^{\widetilde{z},t} = \frac{N_d \boldsymbol{\tau}^{u^{\text{post}},t}}{\text{tr}((\mathbf{H}^{-1})^H\mathbf{H}^{-1})}. \tag{47}$$

The decoupling-coupling transformation exploits the channel matrix to disperse clipping distortion, which effectively enhances the algorithm performance. The forward messages are approximated by Taylor expansion as

$$\mu_{z_n\to x_m}^t(x_m) \approx 2\Re\{s_n^t A_{n,m}^*\left(x_m - \hat{x}_m^{\text{post},t}\right)\} - \tau_n^{s,t}\left|A_{n,m}\right|^2\left|x_m - \hat{x}_m^{\text{post},t}\right|^2, \tag{48}$$

where the *a priori* messages $\mathbf{z}^{\text{prior}}$ (i.e., $\mathbf{s}^t$ and $\boldsymbol{\tau}^{s,t}$) are represented as

$$s_n^t = \frac{\widetilde{z}_n^t - \hat{p}_n^t}{\tau_n^{\hat{p},t}}, \tag{49}$$

$$\tau_n^{s,t} = \frac{\tau_n^{\hat{p},t} - \tau_n^{\widetilde{z},t}}{\left|\tau_n^{\hat{p},t}\right|^2}. \tag{50}$$

Furthermore, the Taylor expansion of the backward messages can be approximated as

$$\mu_{x_m\to z_n}^{t+1}(x_m) \approx \log p(x_m \mid d_m) - \frac{|\hat{r}_{x_m\to z_n}^t - x_m|^2}{\tau_{x_m\to z_n}^{r,t}}, \tag{51}$$

where

$$\tau_{x_m\to z_n}^{r,t} = \left(\sum_{n'\neq n}\left|A_{n',m}\right|^2 \tau_{n'}^{s,t}\right)^{-1}, \tag{52}$$

$$\hat{r}_{x_m\to z_n}^t = \tau_{x_m\to z_n}^{r,t}\sum_{n'\neq n}\left[s_{n'}^t A_{n',m}^* + \left|A_{n',m}\right|^2 \tau_{n'}^{s,t}\hat{x}_m^{\text{post},t}\right] = \hat{x}_m^{\text{post},t} + \tau_{x_m\to z_n}^{r,t}\sum_{n'\neq n} s_{n'}^t A_{n',m}^*, \tag{53}$$

and the likelihood messages $\mathbf{x}^{\text{like}}$ (i.e., $\hat{r}_{x_m}^t$ and $\tau_{x_m}^{r,t}$) are represented as

$$\tau_{x_m}^{r,t} = \left(\sum_n \left|A_{n,m}\right|^2 \tau_n^{s,t}\right)^{-1}, \tag{54}$$

$$\hat{r}_{x_m}^t = \tau_{x_m}^{r,t}\sum_n\left[s_n^t A_{n,m}^* + \left|A_{n,m}\right|^2 \tau_n^{s,t}\hat{x}_m^{\text{post},t}\right] = \hat{x}_m^{\text{post},t} + \tau_{x_m}^{r,t}\sum_n s_n^t A_{n,m}^*. \tag{55}$$

According to the aforementioned derivations, the VMGAMP and rVMGAMP symbol detection algorithms for clipped OCDM systems are summarized in **Algorithm** 2.

It is worth clarifying that rVMGAMP is an ICG-based refinement of VMGAMP. Their only difference lies in the nonlinear posterior update for a sample detected as clipped. VMGAMP approximates the aggregate effective error by a CSCG variable and therefore characterizes its uncertainty using only a scalar variance. This provides a low-complexity approximation when clipping is mild or moderate, or when the effective error is dominated by AWGN. In contrast, signal clipping limits the signal magnitude while preserving its phase. After channel decoupling, the resulting clipping error is therefore predominantly distributed along the radial direction under severe-clipping and high-SNR conditions, for which the circularly symmetric approximation becomes inaccurate. Accordingly, rVMGAMP retains the pseudo-covariance and applies the ICG posterior updates in (17)–(20) to capture this second-order noncircularity. Therefore, VMGAMP provides a lower-complexity option, whereas rVMGAMP provides improved robustness under severe clipping by using a more accurate second-order distortion model.

D. CEP-VMGAMP-Based Joint Channel Estimation and Symbol Detection Algorithm

The message exchange among the channel estimator, the symbol detector, and the LDPC decoder is described in this subsection. At the first turbo iteration, the decoder extrinsic LLRs are initialized as $L_{\text{dec}}^{e,0}(c_m^q) = 0$. The channel feedback messages are initialized by the pilot-based Gaussian prior, i.e., $m_{h_l\to f_n}^0 = m_{h_l}^a$ and $v_{h_l\to f_n}^0 = v_{h_l}^a$.

The *a priori* probability of data bits is expressed as [26]

$$p\left(c_m^q\right) = \frac{1}{2}\left[1 + (-1)^{c_m^q}\tanh\left(\frac{1}{2}L_{dec}^{e,t}\left(c_m^q\right)\right)\right], \tag{56}$$

where $L_{\text{dec}}^{e,t}\left(c_m^q\right)$ is the *a priori* extrinsic log-likelihood ratio (LLR), c_m^q is the q-th bit of x_m, and the *a priori* probability of $\mathbf{x}$ is expressed as

$$p(\hat{x}_m^{a,t} = \mathcal{M}_{m_c}) = \prod_q p(c_m^q = c_{m_c}^q). \tag{57}$$

The likelihood probability of $\mathbf{x}$ is expressed as

$$p(\hat{x}_m^{\text{like},t} = \mathcal{M}_{m_c}) \propto \exp\left(-\frac{\left|\mathcal{M}_{m_c} - \hat{r}_{x_m}^{t-1}\right|^2}{\tau_{x_m}^{r,t-1}}\right), \tag{58}$$

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and the *a posteriori* probability of $\mathbf{x}$ is $p(\hat{x}_m^{\text{post},t+1} = \mathcal{M}_{m_c}) \propto p(\hat{x}_m^{a,t} = \mathcal{M}_{m_c}) p(\hat{x}_m^{\text{like},t} = \mathcal{M}_{m_c})$. The *a posteriori* mean and variance of $\mathbf{x}$ are expressed as

$$\hat{x}_m^{\text{post},t+1} \propto \sum_{m_c} \mathcal{M}_{m_c} p(\hat{x}_m^{a,t} = \mathcal{M}_{m_c}) p(\hat{x}_m^{\text{like},t} = \mathcal{M}_{m_c}), \tag{59}$$

$$\hat{\tau}_{x_m}^{\text{post},t+1} \propto \sum_{m_c} |\hat{x}_m^{\text{post},t+1} - \mathcal{M}_{m_c}|^2 p(\hat{x}_m^{a,t} = \mathcal{M}_{m_c}) \times p(\hat{x}_m^{\text{like},t} = \mathcal{M}_{m_c}). \tag{60}$$

Further, the *a posteriori* message $\mathbf{x}^{\text{post}}$ is transmitted to channel estimator and symbol detector, respectively.

After obtaining the likelihood messages of the channel parameters, the *a posteriori* messages of the channel parameters are calculated by

$$m_{h_l}^{\text{post},t} = v_{h_l}^t \left(\frac{m_{h_l}^a}{v_{h_l}^a} + \sum_{n_{\text{blk}}} \sum_n \Xi_{n,l}^* \frac{\hat{b}_{f_n \to h_l}^{t,n_{\text{blk}}}}{\hat{\tau}_{f_n \to h_l}^{t,n_{\text{blk}}}} \right), \tag{61}$$

$$v_{h_l}^{\text{post},t} = \left((v_{h_l}^a)^{-1} + \sum_{n_{\text{blk}}} \sum_n (\hat{\tau}_{f_n \to h_l}^{t,n_{\text{blk}}})^{-1} \right)^{-1}, \tag{62}$$

where $n_{\text{blk}} \in [1, N_{\text{blk}}]$. After that, the VMGAMP symbol detection algorithm shown in **Algorithm** 2 is performed to obtain the likelihood messages $\mathbf{x}^{\text{like}}$. Then, the LLR is calculated as

$$L^{\text{in}}(c_m^q) = \ln \frac{p(c_m^q = 0 \mid \hat{r}_{x_m}^t)}{p(c_m^q = 1 \mid \hat{r}_{x_m}^t)} - L_{\text{dec}}^{e,t-1}(c_m^q). \tag{63}$$

The likelihood LLR is fed into the LDPC decoder to obtain the *a priori* extrinsic LLR $L_{\text{dec}}^{e,t}(c_m^q)$, which is then fed back to the next iteration until the iterative algorithm reaches the maximum number of iterations $T_{\max}$ or satisfies the stopping criterion. The *a priori* extrinsic LLR $L_{\text{dec}}^{e,t}(c_m^q)$ is expressed as

$$L_{\text{dec}}^{e,t}(c_m^q) = L^{\text{out}}(c_m^q) - L^{\text{in}}(c_m^q), \tag{64}$$

and convergence factor can be calculated by

$$\eta = \frac{1}{N_c} \sum_{n=1}^{N_{\text{blk}}} \sum_{m=1}^{N_d} \sum_{q=1}^{Q} \mathbb{I}\left(|L^{\text{out}}(c_m^q)| > 10 \right), \tag{65}$$

where $\mathbb{I}$ is the indicator function.

Based on the aforementioned derivations, the CEP-VMGAMP-based joint channel estimation and symbol detection algorithm for clipped OCDM signal is summarized in **Algorithm** 3.

E. Complexity Analysis

Since soft information computation and BP decoding are standard procedures in LDPC-coded systems, we only consider the computational complexity of channel estimation and symbol detection. Following [28], computational complexity is characterized using floating-point operation (FLOP)-based Big-O notation. An $M \times N$ matrix-vector product has complexity $\mathcal{O}(MN)$, while a length-N vector inner product or element-wise product has complexity $\mathcal{O}(N)$. An N-point FFT/IFFT has complexity $\mathcal{O}(N \log N)$.

Algorithm 1 Proposed CEP Channel Estimation

Initialization: $t = 1, m_{h_l \to f_n}^0 = m_{h_l}^a, v_{h_l \to f_n}^0 = v_{h_l}^a, l = 1, \ldots, L, n = 1, \ldots, N_d$.
Input: $\mathbf{y}$; pilot-initialized channel prior $\mathbf{h}_{\text{TD}}^{\text{prior}}$; symbol posterior messages $\mathbf{x}^{\text{post},t-1}$ from the symbol detector and LDPC decoder; CEP feedback messages $m_{h_l \to f_n}^{t-1}, v_{h_l \to f_n}^{t-1}$; C_r, σ_n^2, N_d, L, and the number of quadrature points K.
for $n = 1$ to N_d **do**
  Compute $\check{b}_{f_n}^t$ and $\tau_{f_n}^t$ with (88) and (89);
  Compute $\hat{m}_{\tilde{u}_n}^t$ and $\hat{v}_{\tilde{u}_n}^t$ with (91) and (92);
  Compute ρ_n^t with (97);
  for $l = 1$ to L **do**
    Compute $\hat{b}_{f_n \to h_l}^t$ and $\hat{\tau}_{f_n \to h_l}^t$ with (25) and (26);
  end for
end for
for $l = 1$ to L **do**
  Compute $m_{h_l}^t$ and $v_{h_l}^t$ with (28) and (29);
  for $n = 1$ to N_d **do**
    Compute $m_{h_l \to f_n}^t$ and $v_{h_l \to f_n}^t$ with (31) and (32);
  end for
end for
Output: Channel likelihood messages $\mathbf{h}_{\text{FD}}^{\text{like},t}$; posterior channel messages $\mathbf{h}_{\text{TD}}^{\text{post},t}$ for Algorithm 2; feedback messages $\mathbf{h}_{\text{TD},b}^{\text{like},t}$ for the CEP update.

Since the DFnT/IDFnT and the application of the circulant channel matrix $\mathbf{H}$ or $\mathbf{H}^{-1}$ can be implemented using FFT/IFFT and element-wise operations, they also have complexity $\mathcal{O}(N \log N)$.

For Algorithm 1, the discrete Fourier transform (DFT)/DFnT operations, channel message updates, and truncated quadrature require $\mathcal{O}(N_d \log N_d)$, $\mathcal{O}(N_d L)$, and $\mathcal{O}(N_d K)$ operations, respectively. Therefore, the per-block complexity of CEP is $\mathcal{O}(N_d \log N_d + N_d L + N_d K)$, where K is the number of sampling points in the truncated adaptive uniform-grid quadrature. For Algorithm 2, the matrix-vector products involving $\mathbf{A}$ and $|\mathbf{A}|^2$ require $\mathcal{O}(N_d^2)$ operations, while the FFT-based channel decoupling and coupling require $\mathcal{O}(N_d \log N_d)$ operations. Hence, VMGAMP and rVMGAMP have complexities $\mathcal{O}(N_d^2 + N_d \log N_d)$ and $\mathcal{O}(N_d^2 + N_d \log N_d + N_{\text{clip}})$, respectively. Since $N_{\text{clip}} \leq N_d$, rVMGAMP has the same dominant complexity order as VMGAMP, where N_{clip} denotes the number of clipped samples.

For a frame with N_{blk} data blocks and T turbo iterations, the overall complexity of the proposed joint receiver is $\mathcal{O}(N_{\text{blk}} N_p \log N_p + T N_{\text{blk}}(N_d^2 + N_d \log N_d + N_d K + N_d L + N_d M_c Q))$, where $M_c = |\mathcal{M}|$ and $Q = \log_2 M_c$. Therefore, the dominant detector complexity scales quadratically with the OCDM block length, whereas the LLR calculations scale as $\mathcal{O}(N_d M_c Q)$.

IV. Cramér–Rao Lower Bound Analysis

To evaluate the performance of the proposed CEP channel estimation algorithm, a CRLB for channel estimation in

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Algorithm 2 Proposed (r)VMGAMP Symbol Detection

Initialization: $t = 1, \hat{x}_m^0 = 0, \tau_m^{x,0} = +\infty, n = 1,\ldots,N_d, m = 1,\ldots,N_d$, $s_n^0 = 0$.
Input: $\mathbf{y}$; posterior channel messages $\mathbf{h}_{\mathrm{TD}}^{\mathrm{post},t}$ from Algorithm 1; symbol posterior messages $\mathbf{x}^{\mathrm{post},t}$; decoder extrinsic LLRs $L_{\mathrm{dec}}^{e,t-1}(c_m^q)$; feedback messages s_n^{t-1}; C_r, σ_n^2.
for $n = 1$ to N_d **do**
  Compute $\hat{p}_n^t$ and $\tau_n^{p,t}$ with (35) and (36);
  Compute $\breve{p}_n^t$ and $\tau_n^{\breve{p},t}$ with (37) and (38);
  Compute $\bar{u}_n^t$ and $\tau_n^{\bar{u}^t}$ with (39) and (40);
  if VMGAMP **then**
    Compute $u_n^{\mathrm{post},t}$ and $\tau_n^{u^{\mathrm{post}},t}$ with (43), (44) and (45);
  end if
  if rVMGAMP **then**
    if $|\breve{p}_n^t| \le C_r$ **then**
      Compute $u_n^{\mathrm{post},t}$ and $\tau_n^{u^{\mathrm{post}},t}$ with (43) and (44);
    else if $|\breve{p}_n^t| > C_r$ **then**
      Compute $u_n^{\mathrm{post},t}$ and $\tau_n^{u^{\mathrm{post}},t}$ with (17)–(20);
    end if
  end if
  Compute $\widetilde{z}_n^t$ and $\tau_n^{\widetilde{z},t}$ with (46) and (47);
  Compute s_n^t and $\tau_n^{s,t}$ with (49) and (50);
end for
for $m = 1$ to N_d **do**
  Compute $\hat{r}_{x_m}^t$ and $\tau_{x_m}^{r,t}$ with (55) and (54);
end for
Output: Symbol likelihood messages $\mathbf{x}^{\mathrm{like},t}$ for LLR calculation and LDPC decoding; posterior time-domain messages $\mathbf{u}^{\mathrm{post},t}$; posterior channel-output messages $\mathbf{z}^{\mathrm{post},t}$; updated residual messages s_n^t.

Algorithm 3 Proposed CEP-VMGAMP Algorithm

Initialization:
$t = 1, \hat{x}_m^0 = \hat{r}_{x_m}^0 = 0, \tau_m^{x,0} = \tau_{x_m}^{r,0} = +\infty, m_{h_l \to f_n}^0 = m_{h_l}^a, v_{h_l \to f_n}^0 = v_{h_l}^a, \eta = 0, \eta_{\max} = -\infty, \varepsilon = 10^{-2}, l = 1,\ldots,L, n = 1,\ldots,N_d, m = 1,\ldots,N_d$.
for $t = 1$ to $T_{\max}$ **do**
  Compute the *a posteriori* mean and variance of $\mathbf{x}$ according to (56), (57), (58), (59) and (60);
  if $t = 1$ **then**
    Calculate channel parameters *a priori* messages according to (21) and (22);
  else if $t > 1$ **then**
    Execute **Algorithm** 1;
    Calculate channel parameters *a posteriori* messages according to (61) and (62);
  end if
  Execute **Algorithm** 2;
  Compute the likelihood LLR according to (63);
  Execute LDPC decoding;
  Compute the extrinsic information according to (64);
  Compute η according to (65);
  if $\eta > \eta_{\max}$ **then**
    $\eta_{\max} = \eta$;
    $L_{\mathrm{old}}^{\mathrm{out}}(c_m^q) = L^{\mathrm{out}}(c_m^q)$;
  else if $\eta < \eta_{\max} - \varepsilon$ **then**
    $L^{\mathrm{out}}(c_m^q) = L_{\mathrm{old}}^{\mathrm{out}}(c_m^q)$;
    break
  end if
end for
return: $L^{\mathrm{out}}(c_m^q)$.

clipped OCDM systems is derived. It should be emphasized that the CRLB derived in this section is a conditional benchmark for channel estimation. Specifically, the transmitted data symbols used to construct the equivalent observation matrix are treated as known conditioning variables when deriving the channel-related Fisher information. This setting isolates the effects of AWGN and clipping-induced signal power reduction on channel estimation. The derivation is consistent with the signal clipping model and the AWGN assumption adopted in the system model, while the CSCG/ICG models in the proposed VMGAMP/rVMGAMP detector are only used to obtain tractable Gaussian messages for iterative detection, and are not required as exact statistical assumptions in the CRLB derivation. Therefore, the bound is not intended to be a strict lower bound for the complete practical iterative receiver, where the data symbols and channel are jointly inferred. Instead, it serves as a theoretical reference for evaluating the proposed CEP-based channel estimator and for quantifying the influence of the clipping threshold.

With CIR $\mathbf{h} = [h_1, \cdots, h_L, 0, \ldots, 0]^T$, the received symbol can be expressed as

$$\mathbf{y} = \mathbf{U}\mathbf{\Phi}^H\mathbf{h} + \mathbf{w}. \tag{66}$$

The PDF of the received symbol is expressed as

$$p(\mathbf{y};\mathbf{h}) \propto \exp\left[-(\mathbf{y}-\mathbf{G}\mathbf{h})^H \Upsilon^{-1}(\mathbf{y}-\mathbf{G}\mathbf{h})\right], \tag{67}$$

where $\mathbf{G} = \mathbf{U}\mathbf{\Phi}^H$, $\Upsilon = \sigma_n^2\mathbf{I}_{N_d}$, and $\mathbf{U}$ is a column-wise circulant matrix whose first column is $\mathbf{x}$.

To evaluate the impact of clipping distortion on channel estimation, the CRLB with clipped signal is derived. By exploiting the circulant structure of $\mathbf{H}$, the likelihood can be equivalently rewritten with respect to the clipped signal $\bar{\mathbf{u}}$, which yields

$$\begin{aligned}&\int -\left(\mathbf{y}-\widetilde{\mathbf{G}}\bar{\mathbf{u}}\right)^H \Upsilon^{-1}\left(\mathbf{y}-\widetilde{\mathbf{G}}\bar{\mathbf{u}}\right)p(\bar{\mathbf{u}})d\bar{\mathbf{u}}\\ &= -|\mathbf{y}|^2\Upsilon^{-1} - \int_0^{\infty}|\widetilde{\mathbf{G}}|^2|\bar{\mathbf{u}}|^2\Upsilon^{-1}p(|\bar{\mathbf{u}}|)d|\bar{\mathbf{u}}|,\end{aligned} \tag{68}$$

where $\widetilde{\mathbf{G}} = \mathbf{H}$. Considering the properties of multi-carrier signal, the independent and identically distributed (i.i.d.) Rayleigh random variables PDF of OCDM time-domain symbol envelope is $p(|\bar{u}|) = 2|\bar{u}|e^{-|\bar{u}|^2}$ [30]. Thus, the power reduction induced by signal clipping can be calculated as

$$\begin{aligned}L_c &= \int_{C_r}^{\infty}(|\bar{u}|^2 - C_r^2)2|\bar{u}|e^{-|\bar{u}|^2}d|\bar{u}|\\ &= [C_r^2e^{-|\bar{u}|^2} - |\bar{u}|^2e^{-|\bar{u}|^2} - e^{-|\bar{u}|^2}]|_{C_r}^{\infty}\\ &= e^{-C_r^2},\end{aligned} \tag{69}$$

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and the second term of (68) can be expressed as

$$-\int_0^{\infty} |\widetilde{\mathbf{G}}|^2|\bar{\mathbf{u}}|^2\Upsilon^{-1}p(|\bar{\mathbf{u}}|)d|\bar{\mathbf{u}}| = -|\widetilde{\mathbf{G}}|^2\Upsilon^{-1}\left(1-e^{-C_r^2}\right). \quad (70)$$

The term $L_c = e^{-C_r^2}$ represents the average signal power loss caused by the signal clipping, and thus $1 - L_c$ represents the average retained signal power after clipping. Accordingly, the clipped-signal Fisher information matrix (FIM) is reduced by the factor $1 - e^{-C_r^2}$.

Considering the power reduction induced by signal clipping, the corresponding FIM is given by [29]

$$\bar{\mathbf{I}}(\mathbf{h}) = \bar{\mathbf{G}}^H\Upsilon^{-1}\bar{\mathbf{G}}, \quad (71)$$

where $\bar{\mathbf{G}}^H\bar{\mathbf{G}} = (1-e^{-C_r^2})\mathbf{G}^H\mathbf{G}$.

Since only the first L entries of the zero-padded CIR vector $\mathbf{h}$ are unknown, let $\mathbf{h}_L = [h_1, \ldots, h_L]^T$, $\mathbf{h} = \mathbf{S}_d\mathbf{h}_L$, where $\mathbf{S}_d = \begin{bmatrix} \mathbf{I}_L \\ \mathbf{0}_{(N_d-L)\times L} \end{bmatrix}$. Accordingly, the data-block FIM associated with the unknown L-tap CIR is obtained by restricting the full FIM to the known channel-support subspace, i.e., $\mathbf{I}_d(\mathbf{h}_L) = \mathbf{S}_d^H\bar{\mathbf{I}}(\mathbf{h})\mathbf{S}_d$. Although $\bar{\mathbf{I}}(\mathbf{h})$ is an $N_d \times N_d$ matrix, $\mathbf{I}_d(\mathbf{h}_L)$ is an $L \times L$ matrix defined with respect to the actual unknown channel parameters. The pilot likelihood is

$$p(\mathbf{y}_p;\mathbf{h}) \propto \exp\left[-(\mathbf{y}_p - \mathbf{G}_p\mathbf{h})^H\boldsymbol{\Upsilon}_p^{-1}(\mathbf{y}_p - \mathbf{G}_p\mathbf{h})\right], \quad (72)$$

where $\boldsymbol{\Upsilon}_p = \sigma_n^2\mathbf{I}_{N_p}$. The FIM contributed by the pilot is

$$\mathbf{I}_p(\mathbf{h}) = \mathbf{G}_p^H\boldsymbol{\Upsilon}_p^{-1}\mathbf{G}_p. \quad (73)$$

Since $\boldsymbol{\Phi}_p$ is unitary, it follows that

$$\mathbf{I}_p(\mathbf{h}_L) = \frac{N_p}{\sigma_n^2}\mathbf{I}_L. \quad (74)$$

Since the pilot and data noise vectors are independent, their conditional likelihoods satisfy

$$p(\mathbf{y}_p, \mathbf{y}; \mathbf{h}) = p(\mathbf{y}_p;\mathbf{h})p(\mathbf{y};\mathbf{h}), \quad (75)$$

and hence the corresponding FIMs are additive. The total channel-related FIM of a frame is

$$\mathbf{I}_{\text{tot}}(\mathbf{h}_L) = N_{\text{blk}}(\mathbf{I}_p(\mathbf{h}_L) + \mathbf{I}_d(\mathbf{h}_L)). \quad (76)$$

Accordingly, the channel estimation mean-square error (MSE) is lower bounded by

$$\text{MSE} \geq \text{CRLB}_{p+d} = \text{tr}\left[\frac{(\mathbf{I}_p(\mathbf{h}_L) + \mathbf{I}_d(\mathbf{h}_L))^{-1}}{N_{\text{blk}}}\right]. \quad (77)$$

V. Numerical Results

In this section, numerical results are presented to evaluate the performance of the proposed VMGAMP-based symbol detector, the CEP-based channel estimator, and the CEP-VMGAMP-based joint channel estimation and symbol detection iterative receiver.

To verify the effectiveness of the proposed algorithms, LS [10], linear minimum mean square error (LMMSE) [31], orthogonal matching pursuit (OMP) [32], and vector approximate message passing (VAMP) [33], [34] are used as channel estimation baselines, while GAMP [35], [36], FGAMP [24], and memory approximate message passing (MAMP) [37] are used as symbol detection baselines.

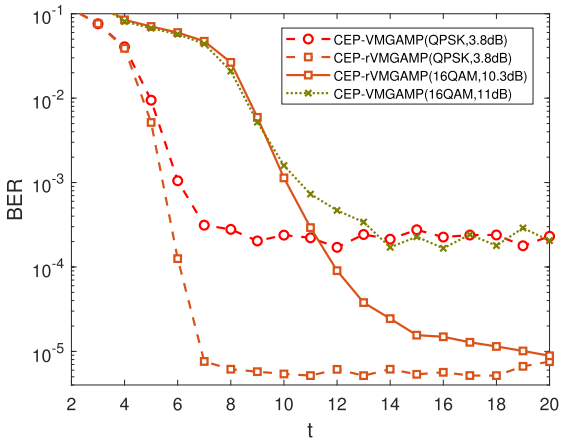


Fig. 5. BER performance of different algorithms with different numbers of turbo iterations for clipped OCDM LDPC-coded systems.

For clarity, clipping VAMP (CVAMP), VAMP-GAMP, and CEP-FGAMP denote the adapted baselines implemented in this paper. Specifically, CVAMP combines VAMP and the clipping approach based on signal reconstruction. Throughout this subsection, the length of CP is 16, which is no shorter than the maximum delay spread. The length of pilot $N_p = 32$, and the number of subcarriers $N_d = 126$. In the LDPC-coded system, an LDPC code with 3/4 code rate (6048/8064) is employed. The maximum number of turbo iterations and LDPC decoding iterations are $T_{\max} = 20$ and $T_{\text{dec}} = 10$, respectively. For the truncated adaptive uniform-grid quadrature, the grid size is fixed as $K = 11 \times 11 = 121$ for all simulated SNRs and clipping thresholds. No damping factor is used in either the proposed algorithms or the baseline algorithms. The number of symbol blocks $N_{\text{blk}} = 32$ for quadrature phase shift keying (QPSK) and $N_{\text{blk}} = 16$ for 16QAM. The channel power delay profile (PDP) follows the Extended Vehicular A (EVA) model [38].

A. Empirical Convergence Behavior

In Fig. 5, the empirical convergence behavior of the proposed CEP-VMGAMP and CEP-rVMGAMP receivers is evaluated in terms of BER versus the turbo-iteration index t for clipped LDPC-coded OCDM systems, where the clipping threshold is set to $C_r = 1.2$. For QPSK, both receivers are evaluated at $\text{E}_\text{b}/\text{N}_0 = 3.8$ dB. For 16QAM, CEP-VMGAMP and CEP-rVMGAMP are evaluated at $\text{E}_\text{b}/\text{N}_0 = 11$ dB and 10.3 dB, respectively. Since the two receivers exhibit substantially different BER operating points for 16QAM, the SNRs are selected separately so that the iteration trajectory of each receiver can be clearly observed.

Under the considered settings, the BER curves generally decrease with the turbo-iteration index and then stabilize, without persistent oscillation or divergence. For QPSK, both receivers reach a stable BER regime within approximately $t = 7$ turbo iterations, indicating that the performance gain of CEP-rVMGAMP is achieved without requiring additional

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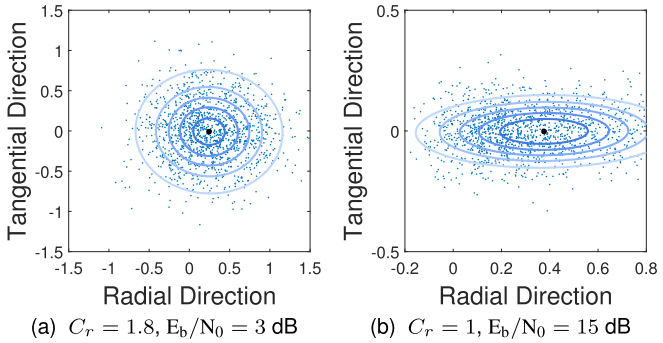


Fig. 6. Scatter plots of the phase-aligned effective-error samples. Each sample is rotated according to the corresponding phase used in the ICG update, so that the horizontal and vertical axes represent the radial and tangential components, respectively.

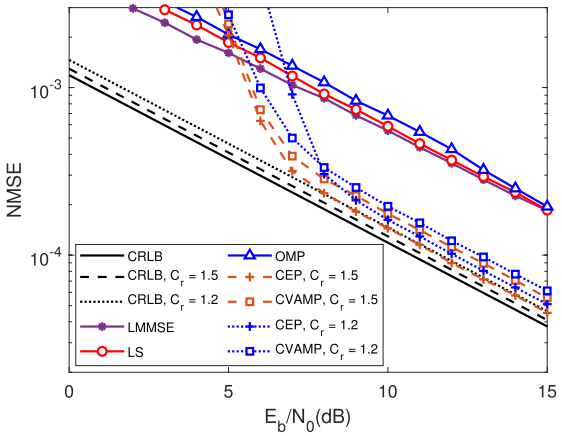


Fig. 7. NMSE performance of different channel estimation algorithms for QPSK OCDM uncoded systems.

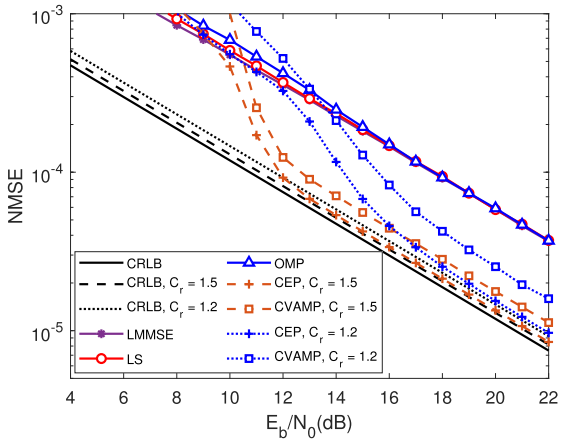


Fig. 8. NMSE performance of different channel estimation algorithms for 16QAM OCDM uncoded systems.

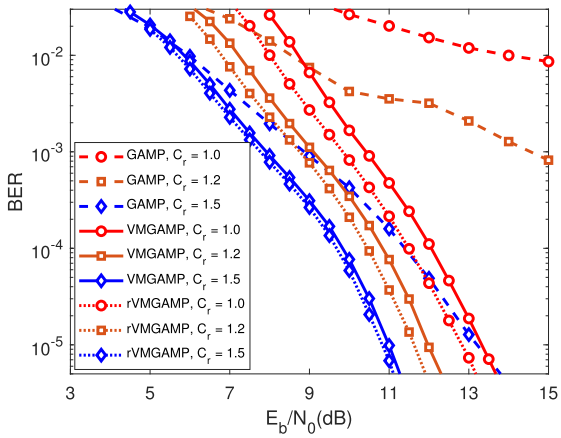


Fig. 9. BER performance for QPSK OCDM uncoded systems with known CSI and different clipping thresholds.

turbo iterations at the same SNR. For 16QAM, each receiver reaches a stable BER regime within approximately $t = 15$ turbo iterations. These results provide empirical evidence of stable iterative behavior for the proposed receivers under the evaluated settings.

B. Empirical Validation of the CSCG/ICG Approximations

Fig. 6 empirically examines the second-order structure of the effective error. Under mild clipping and low E_b/N_0, the radial and tangential components have comparable spreads, for which the scalar variance used by VMGAMP provides a reasonable low-complexity approximation. Under severe clipping and high E_b/N_0, the pronounced radial elongation results in nonzero pseudo-covariance, which motivates the ICG-based rVMGAMP update.

C. NMSE Performance

Fig. 7 and Fig. 8 present the normalized mean square error (NMSE) performance of the proposed CEP-based channel estimator in an uncoded system, and the CRLBs corresponding to different clipping thresholds are also provided as benchmarks. For a fair comparison, each data-aided channel estimation method incorporates the clipping approach based on signal reconstruction and adopts the VMGAMP symbol detection algorithm. Since the CEP algorithm performs moment matching on the frequency-domain symbols, it achieves better NMSE than the CVAMP algorithm. It is noted that the plotted CRLB is the conditional channel estimation benchmark derived in Section IV, rather than the attainable limit of the complete joint receiver. Hence, the CEP estimator yields NMSE values slightly above this bound, since the data symbols are unknown in practice, the channel is refined from soft decisions within finite turbo iterations, and the moment matching required by the CEP update is evaluated through quadrature-based approximate marginalization. Moreover, a smaller C_r gives a larger $L_c = e^{-C_r^2}$, reduces the effective Fisher information, and thus leads to a higher CRLB, which is consistent with the degradation caused by stronger clipping.

D. BER Performance

The bit error rate (BER) performance of the symbol detector with known channel state information (CSI) is evaluated in Figs. 9–12. In the uncoded QPSK systems shown in Fig. 9, the proposed VMGAMP and rVMGAMP detectors achieve lower BER than the GAMP under the considered clipping thresholds. In particular, the proposed VMGAMP algorithm provides a 2.3 dB E_b/N_0 gain over GAMP at BER $= 10^{-5}$ when

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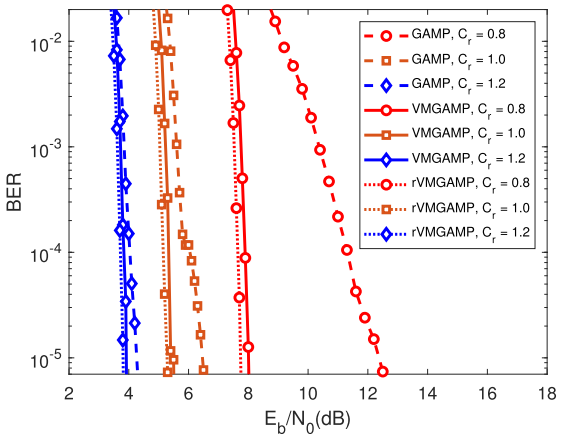


Fig. 10. BER performance for QPSK OCDM LDPC-coded systems with known CSI and different clipping thresholds.

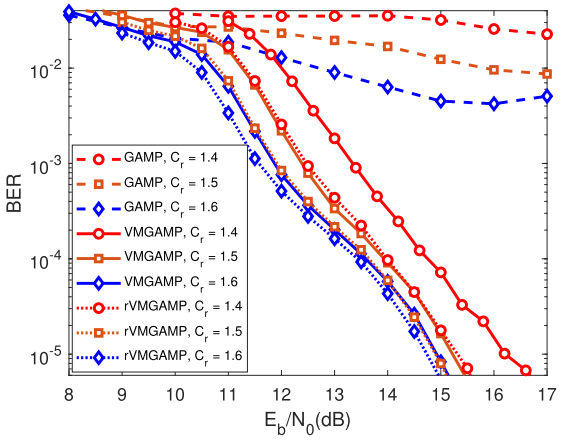


Fig. 11. BER performance for 16QAM OCDM uncoded systems with known CSI and different clipping thresholds.

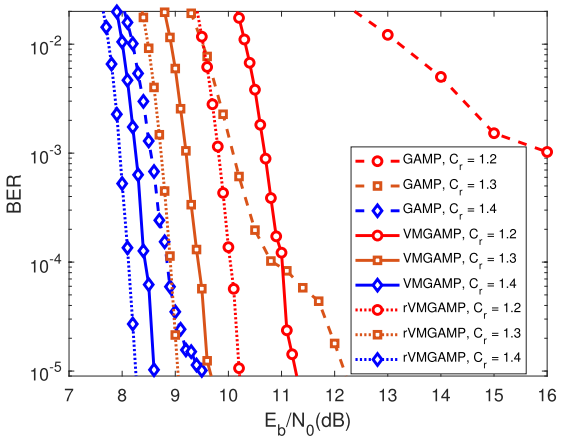


Fig. 12. BER performance for 16QAM OCDM LDPC-coded systems with known CSI and different clipping thresholds.

$C_r = 1.5$, and the GAMP algorithm fails under more severe clipping. In the LDPC-coded QPSK systems, the proposed VMGAMP algorithm provides 0.4 dB, 1 dB, and 4.5 dB E_b/N_0 gains over GAMP at BER $= 10^{-5}$ when $C_r = 1.2$, $C_r = 1.0$, and $C_r = 0.8$, respectively.

As shown in Fig. 11, the proposed VMGAMP and rVMGAMP detectors can effectively mitigate clipping distortion and achieve reliable detection in uncoded 16QAM

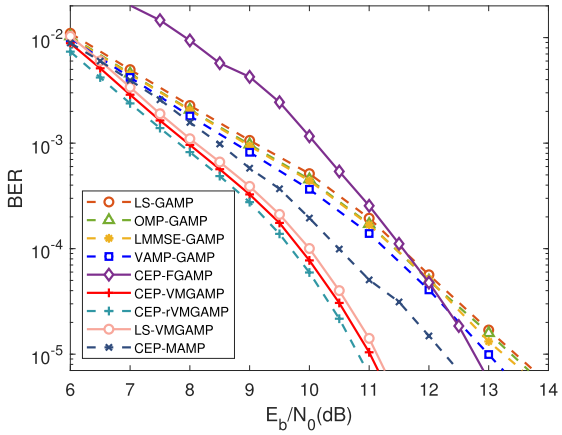


Fig. 13. BER performance of different joint channel estimation and symbol detection algorithms for QPSK OCDM uncoded systems.

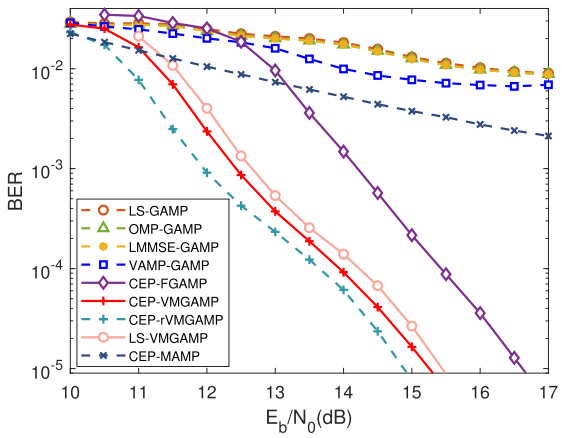


Fig. 14. BER performance of different joint channel estimation and symbol detection algorithms for 16QAM OCDM uncoded systems.

systems. As shown in Fig. 12, in the LDPC-coded 16QAM systems, the proposed VMGAMP algorithm provides 0.9 dB and 2.5 dB E_b/N_0 gains over GAMP at BER $= 10^{-5}$ when $C_r = 1.4$ and $C_r = 1.3$, respectively. Moreover, the GAMP algorithm fails to handle more severe clipping. It is worth noting that the comparison between QPSK and 16QAM demonstrates that higher-order modulation schemes are more sensitive to clipping distortion. Overall, these results validate the effectiveness of the VMGAMP algorithm, particularly when the higher-order modulation schemes are employed. Moreover, the performance advantage of rVMGAMP algorithm becomes more evident as the clipping distortion intensifies.

In Fig. 13 and Fig. 14, the BER performance of the proposed CEP-VMGAMP-based joint channel estimation and symbol detection algorithm for uncoded systems is evaluated. The clipping threshold is $C_r = 1.5$. For QPSK systems, CEP-VMGAMP provides more than 1.8 dB E_b/N_0 gain over the LS-GAMP, LMMSE-GAMP, OMP-GAMP, and VAMP-GAMP receiver baselines at BER $= 10^{-5}$. The ablation study comparing LS-VMGAMP with CEP-VMGAMP validates the effectiveness of the CEP-based data-aided channel estimation. The complementary ablation study comparing LS-GAMP

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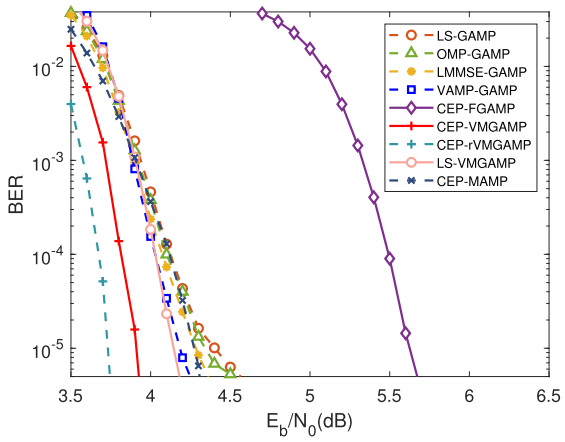


Fig. 15. BER performance of different joint channel estimation and symbol detection algorithms for QPSK OCDM LDPC-coded systems.

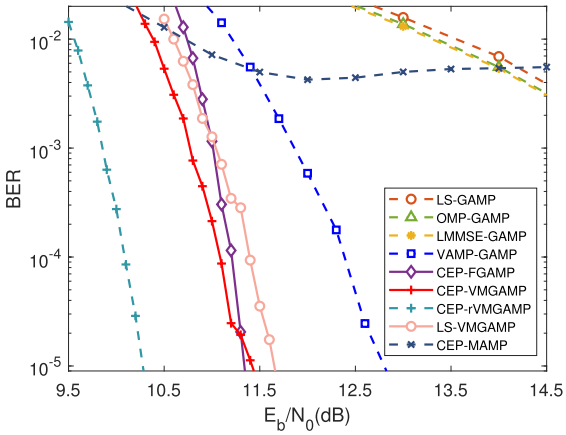


Fig. 16. BER performance of different joint channel estimation and symbol detection algorithms for 16QAM OCDM LDPC-coded systems.

with LS-VMGAMP validates the combined effectiveness of the channel decoupling-coupling transformation and variance modification introduced in VMGAMP. For 16QAM, the symbol detection baselines without explicit clipping distortion mitigation exhibit substantial performance degradation. For comparison, CEP-VMGAMP provides a 1.4 dB E_b/N_0 gain over CEP-FGAMP at BER $= 10^{-5}$.

The BER performance of the proposed CEP-VMGAMP-based joint channel estimation and symbol detection algorithm for the LDPC-coded systems is illustrated in Fig. 15 and Fig. 16. The clipping threshold is $C_r = 1.2$. As shown in Fig. 15, the proposed CEP-VMGAMP algorithm outperforms the comparison algorithms without explicit clipping distortion mitigation by more than 0.3 dB at BER $= 10^{-5}$ when QPSK is adopted. In contrast, LS-VMGAMP remains effective, and the performance gain of CEP-VMGAMP over LS-VMGAMP demonstrates the benefit of the CEP-based data-aided channel estimation. Furthermore, it provides a 1.8 dB E_b/N_0 gain over CEP-FGAMP at BER $= 10^{-5}$. As shown in Fig. 16, the proposed CEP-VMGAMP algorithm provides a 1.3 dB E_b/N_0 gain over VAMP-GAMP at BER $= 10^{-5}$ when 16QAM is adopted. Moreover, with $C_r = 1.2$, the BER performance of 16QAM is dominated by clipping distortion. Consequently, the VMGAMP and FGAMP algorithms deliver similar performance under this condition. The numerical results indicate that the proposed CEP-VMGAMP algorithm outperforms the compared baselines. Moreover, the simulation results indicate that the rVMGAMP algorithm achieves more pronounced performance gains with higher-order constellations, since higher-order modulation is more sensitive to clipping distortion.

Since LS-VMGAMP and CEP-VMGAMP employ the same pilot configuration and the same VMGAMP detector, their BER difference isolates the benefit of the CEP-based data-aided channel estimation. Likewise, since CEP-FGAMP and CEP-VMGAMP use the same CEP estimator, their performance difference reflects the gain obtained from the channel decoupling-coupling and variance modification operations of VMGAMP. Similarly, CEP-MAMP, CEP-VMGAMP, and CEP-rVMGAMP employ the same CEP-based channel estimation algorithm and pilot configuration. The lower BER of CEP-VMGAMP and CEP-rVMGAMP relative to CEP-MAMP, particularly for 16QAM, supports the benefit of explicitly incorporating channel decoupling-coupling, clipping detection, and variance modification. The rVMGAMP detector further improves robustness under severe clipping by retaining the clipping-induced pseudo-covariance through the ICG-based posterior update. The CEP-related overhead is therefore more valuable when the pilot-only channel estimate is accuracy-limited, such as under fixed and limited pilot overhead or in the low-to-moderate E_b/N_0 region, whereas the additional VMGAMP/rVMGAMP processing becomes more beneficial as the clipping distortion becomes stronger or the modulation order increases.

VI. Conclusion

In this paper, we proposed an iterative receiver for clipped OCDM systems consisting of a CEP-based channel estimator and a VMGAMP-based symbol detector. By analyzing the properties of clipping distortion, a factor graph was constructed by introducing a gate mechanism. The CEP-based channel estimator incorporated the clipping approach based on signal reconstruction and truncated adaptive uniform-grid quadrature to improve estimation accuracy and reduce computational complexity. Meanwhile, the VMGAMP-based symbol detector employs a decoupling-coupling transformation and variance modification to mitigate clipping distortion. In particular, within the proposed VMGAMP framework, the ICG model achieves better detection performance under severe clipping at the cost of a slight increase in computational complexity. To provide a benchmark, we derive a closed-form CRLB for channel estimation in clipped OCDM systems. The proposed receiver consistently improves the channel estimation NMSE and the BER over the compared baselines, with more pronounced gains under more severe clipping and higher-order modulation.

Appendix A
Detailed BP and ICG Message Updates

This appendix provides the BP equations omitted from Section III and the detailed derivation used in the ICG message

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update. According to BP rules [26], the message updating is expressed as

$$\mu_{n\to m}(a_m) \propto \int_{\mathbf{a}\setminus a_m} f_n(\mathbf{a}) \prod_{m'\in\mathcal{N}(n)\setminus m} \mu_{n\leftarrow m'}(a_{m'}), \quad (78)$$

$$\mu_{n\leftarrow m}(a_m) \propto \prod_{n'\in\mathcal{N}(m)\setminus n} \mu_{n'\to m}(a_m), \quad (79)$$

where f_n is factor node, a_m is variable node, $\mathcal{N}(m)\setminus n$ is the set of neighbors of node m excluding node n. For clipping node $p(\bar{\mathbf{u}} \mid \mathbf{u})$, the message updating is

$$\mu_{f_n\to u_n} \propto \int_{\bar{u}_n} \delta(u_n - \text{clip}^{-1}(\bar{u}_n))\mu_{f_n\leftarrow\bar{u}_n}, \quad (80)$$

$$\mu_{f_n\leftarrow u_n} \propto \int_{\mathbf{x}} \delta(u_n - \mathbf{\Phi}^H_{n,:}\mathbf{x})p(\mathbf{x}), \quad (81)$$

$$\mu_{f_n\to\bar{u}_n} \propto \int_{u_n} \delta(\bar{u}_n - \text{clip}(u_n))\mu_{f_n\leftarrow u_n}, \quad (82)$$

$$\mu_{f_n\leftarrow\bar{u}_n} \propto \int_{\bar{\mathbf{z}},\mathbf{h}} \delta(\bar{u}_n - \mathbf{H}^{-1}_{n,:}\bar{\mathbf{z}})p(\bar{\mathbf{z}})p(\mathbf{h}). \quad (83)$$

However, since signal clipping is inherently irreversible, the generalized inverse $\text{clip}^{-1}(\bar{u}_n)$ lacks closed-form expression, and thus the message updates are analytically intractable and computationally prohibitive. Moreover, standard nonlinear approximation such as Taylor-series and Volterra-series expansions are ill-suited for clipping because of its clipping nonlinearity. Modeling the clipping distortion as an additive noise term is common, but it may incur substantial performance degradation, particularly when the clipping distortion power is non-negligible. Although existing studies have developed heuristic compensation schemes, these methods typically provide limited robustness and still leave a non-trivial performance gap.

APPENDIX B
ICG UPDATE

For completeness, this subsection provides the covariance derivation used in the ICG message update in Section III-A. The corresponding covariance matrix is

$$\begin{aligned} C &= \mathbb{E}[(u_a-\mu_a)(u_a-\mu_a)^H] \\ &= \begin{pmatrix} \tau_n^{\bar{u},t} + \hat{\tau}^t_{\text{clip},n} & p_n^{\text{var},t} \\ (p_n^{\text{var},t})^* & \tau_n^{\bar{u},t} + \hat{\tau}^t_{\text{clip},n} \end{pmatrix}, \end{aligned} \quad (84)$$

where $u_a \triangleq \begin{pmatrix} u_n \\ u_n^* \end{pmatrix}$, $\mu_a \triangleq \begin{pmatrix} \bar{u}_n^t \\ (\bar{u}_n^t)^* \end{pmatrix}$, p_n^{var} is the pseudo-covariance of ICG distribution of $\bar{u}_n$, which can be calculated as

$$\begin{aligned} p_n^{\text{var},t} &= \mathbb{E}\big[(u_n-\bar{u}_n)(u_n-\bar{u}_n)^T\big] \\ &= \mathbb{E}\big[e^{j2\phi_n}(n_\parallel + jn_\perp)^2\big] \\ &= \hat{\tau}^t_{\text{clip},n} e^{j2\phi^t_n}, \end{aligned} \quad (85)$$

where $\mathbb{E}[n_\parallel^2] = \frac{\tau_n^{\bar{u}}}{2} + \hat{\tau}_{\text{clip},n}$ and $\mathbb{E}[n_\perp^2] = \frac{\tau_n^{\bar{u}}}{2}$ denote the variances of the radial and tangential error components, respectively.

APPENDIX C
DERIVATION OF THE CEP MOMENT-MATCHING UPDATE

According to the BP rules, the likelihood messages of h_l can be calculated as

$$\gamma(h_l) = \int_{\mathbf{h}\setminus h_l}\int_{\widetilde{u}_n} f_n(\widetilde{u}_n,\mathbf{h})\mu_{\widetilde{u}_n\to f_n}(\widetilde{u}_n)\prod_{l'\neq l}\mu^{t-1}_{h_{l'}\to f_n}(h_{l'}), \quad (86)$$

where t is the number of turbo iterations, $\widetilde{\mathbf{u}} = \mathbf{\Gamma}^H\mathbf{F}\bar{\mathbf{x}}$, f_n is the n-th factor node of $p(\bar{\mathbf{z}} \mid \mathbf{h},\bar{\mathbf{u}})$, and the messages of estimation error are expressed as

$$p(\widetilde{\epsilon}) = \int_{\widetilde{u}_n}\mu_{\widetilde{u}_n\to f_n}(\widetilde{u}_n)\,\mathcal{CN}\Big(0;\widetilde{u}_n\check{b}^t_{f_n}(\widetilde{u}_n), |\widetilde{u}_n|^2\tau^t_{f_n}(\widetilde{u}_n)\Big), \quad (87)$$

$$\check{b}^t_{f_n}(\widetilde{u}_n) = \frac{1}{\widetilde{u}_n}\mathbf{f}_0^T(n)\mathbf{y} - \Lambda^{t-1}_{n,n}, \quad (88)$$

$$\tau^t_{f_n}(\widetilde{u}_n) = \frac{\sigma_n^2}{|\widetilde{u}_n|^2} + \sum_l v^{t-1}_{h_l\to f_n}, \quad (89)$$

where $\mathbf{\Lambda^{t-1}} = \mathcal{D}(\mathbf{\Xi}\mathbf{m}^{t-1}_{h\to f})$ is channel frequency response (CFR), $\mathbf{F}$ is a DFT matrix, $\mathbf{\Gamma}^H$ is a coefficient matrix with main diagonal element $e^{-j\pi n^2/N_d}$, and the mean and variance of $\mathbf{h}^{\text{prior}}_{\text{FD}}$ are the first terms of (88) and (89), respectively. The equivalent transformation matrix Ξ is expressed as

$$\begin{bmatrix} \mathbf{f}_0^T(1)\mathbf{f}^{-1}(1) & \cdots & \mathbf{f}_{k-1}^T(1)\mathbf{f}^{-1}(1) & \cdots & \mathbf{f}_{K-1}^T(1)\mathbf{f}^{-1}(1) \\ \vdots & & & & \\ \mathbf{f}_0^T(i)\mathbf{f}^{-1}(i) & & \ddots & & \vdots \\ \vdots & & & & \\ \mathbf{f}_0^T(I)\mathbf{f}^{-1}(I) & \cdots & \mathbf{f}_{k-1}^T(I)\mathbf{f}^{-1}(I) & \cdots & \mathbf{f}_{K-1}^T(I)\mathbf{f}^{-1}(I) \end{bmatrix}$$

where $\mathbf{f}^T_{k-1}(i)$ is the i-th row of DFT matrix with $k-1$ right circular shifts, and $\mathbf{f}^{-1}(i)$ is the i-th column of inverse DFT (IDFT) matrix. According to the central limit theorem (CLT), the *a posteriori* messages $\widetilde{\mathbf{u}}^{\text{post}}$ are approximated by complex Gaussian distribution as

$$\mu^t_{\widetilde{u}_n\to f_n}(\widetilde{u}_n) = \mathcal{CN}(\widetilde{u}_n;\hat{m}^t_{\widetilde{u}_n},\hat{v}^t_{\widetilde{u}_n}), \quad (90)$$

where

$$\hat{m}^t_{\widetilde{u}_n}(\widetilde{u}_n) = \mathbf{F}\cdot\text{clip}\left(\mathbf{\Phi}^H\hat{\mathbf{x}}^{\text{post},t-1}\right), \quad (91)$$

$$\hat{v}^t_{\widetilde{u}_n}(\widetilde{u}_n) = \left(1 - e^{\frac{-C_r{}^2}{1+\sigma_x^2}}\right)\sigma_x^2, \quad (92)$$

where $\mathbf{\Gamma}^H = \mathbf{F}\mathbf{\Phi}^H\mathbf{F}^H$, $\sigma_x^2 = \frac{1}{M}\sum_m \hat{\tau}^{\text{post},t-1}_{x_m}$. The mean and variance of $\Xi_{n,l}h_l$ can be expressed as

$$\begin{aligned} m^t_{\Xi_{n,l}h_l}(\widetilde{u}_n) &= \frac{v^{t-1}_{h_l\to f_n}}{\tau^t_{f_n}(\widetilde{u}_n)}(\check{b}^t_{f_n}(\widetilde{u}_n) + \Xi_{n,l}m^{t-1}_{h_l\to f_n}) \\ &\quad + \left(1 - \frac{v^{t-1}_{h_l\to f_n}}{\tau^t_{f_n}(\widetilde{u}_n)}\right)\Xi_{n,l}m^{t-1}_{h_l\to f_n} \\ &= v^{t-1}_{h_l\to f_n}\frac{\check{b}^t_{f_n}(\widetilde{u}_n)}{\tau^t_{f_n}(\widetilde{u}_n)} + \Xi_{n,l}m^{t-1}_{h_l\to f_n}, \end{aligned} \quad (93)$$

$$v^t_{\Xi_{n,l}h_l}(\widetilde{u}_n) = ((\tau^t_{f_n}(\widetilde{u}_n) - v^{t-1}_{h_l\to f_n})^{-1} + (v^{t-1}_{h_l\to f_n})^{-1})^{-1}$$

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$$= \left(1 - \frac{v^{t-1}_{h_l \to f_n}}{\tau^t_{f_n}(\widetilde{u}_n)}\right) v^{t-1}_{h_l \to f_n}. \quad (94)$$

To achieve an accurate global approximation of the true probability distribution, it is necessary to perform moment matching to minimize the Kullback-Leibler (KL) divergence. The first moment and the second moment of the distribution are given by

$$\hat{m}^t_{\Xi_{n,l}h_l} = \mathbb{E}_{\widetilde{u}_n}\left[m^t_{\Xi_{n,l}h_l}(\widetilde{u}_n)\right] = \Xi_{n,l} m^{t-1}_{h_l \to f_n} + v^{t-1}_{h_l \to f_n} \mathbb{E}_{\widetilde{u}_n}\left[\frac{\check{b}^t_{f_n}(\widetilde{u}_n)}{\tau^t_{f_n}(\widetilde{u}_n)}\right], \quad (95)$$

$$\hat{v}^t_{\Xi_{n,l}h_l} = \mathbb{E}_{\widetilde{u}_n}\left[v^t_{\Xi_{n,l}h_l}(\widetilde{u}_n) + |m^t_{\Xi_{n,l}h_l}(\widetilde{u}_n)|^2\right] - |\hat{m}^t_{\Xi_{n,l}h_l}|^2 = \left(1 - v^{t-1}_{h_l \to f_n}\rho^t_n\right) v^{t-1}_{h_l \to f_n}, \quad (96)$$

where

$$\rho^t_n = \mathbb{E}_{\widetilde{u}_n}\left[\frac{1}{\tau^t_{f_n}(\widetilde{u}_n)} - \left|\frac{\check{b}^t_{f_n}(\widetilde{u}_n)}{\tau^t_{f_n}(\widetilde{u}_n)}\right|^2\right] + \left|\mathbb{E}_{\widetilde{u}_n}\left[\frac{\check{b}^t_{f_n}(\widetilde{u}_n)}{\tau^t_{f_n}(\widetilde{u}_n)}\right]\right|^2. \quad (97)$$

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REFERENCES

[1] X. Ouyang and J. Zhao, "Orthogonal chirp division multiplexing," *IEEE Trans. Commun.*, vol. 64, no. 9, pp. 3946–3957, Sep. 2016.

[2] X. Ouyang, O. A. Dobre, Y. L. Guan, and P. Townsend, "Channel estimation for multiple-input multiple-output orthogonal chirp-division multiplexing systems," *IEEE Trans. Wireless Commun.*, vol. 23, no. 1, pp. 436–449, Jan. 2024.

[3] R. Bomfin, M. Chafii, A. Nimr, and G. Fettweis, "A robust baseband transceiver design for doubly-dispersive channels," *IEEE Trans. Wireless Commun.*, vol. 20, no. 8, pp. 4781–4796, Aug. 2021.

[4] M. S. Omar and X. Ma, "Performance analysis of OCDM for wireless communications," *IEEE Trans. Wireless Commun.*, vol. 20, no. 7, pp. 4032–4043, Jul. 2021.

[5] H. Haif, S. E. Zegrar, and H. Arslan, "Novel OCDM transceiver design for doubly-dispersive channels," *IEEE Trans. Veh. Technol.*, vol. 73, no. 8, pp. 11237–11248, Aug. 2024.

[6] M. Chafii, L. Bariah, S. Muhaidat, and M. Debbah, "Twelve scientific challenges for 6G: Rethinking the foundations of communications theory," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 868–904, 2nd Quart., 2023.

[7] R. Bomfin, M. Chafii, and G. Fettweis, "Low-complexity iterative receiver for orthogonal chirp division multiplexing," in *Proc. IEEE Wireless Commun. Netw. Conf. Workshop (WCNCW)*, Apr. 2019, pp. 1–6.

[8] S. Li et al., "Orthogonal chirp division multiplexing assisted dual-function radar communication in IoT networks," *IEEE Internet Things J.*, vol. 11, no. 13, pp. 23752–23764, Jul. 2024.

[9] B. Wang, Y. Wang, Y. Li, and X. Guan, "Underwater acoustic communications based on OCDM for Internet of Underwater Things," *IEEE Internet Things J.*, vol. 10, no. 24, pp. 22128–22142, Dec. 2023.

[10] B. Wang and X. Guan, "Channel estimation for underwater acoustic communications based on orthogonal chirp division multiplexing," *IEEE Signal Process. Lett.*, vol. 28, pp. 1883–1887, 2021.

[11] X. Ouyang and J. Zhao, "Orthogonal chirp division multiplexing for coherent optical fiber communications," *J. Lightw. Technol.*, vol. 34, no. 18, pp. 4376–4386, Sep. 15, 2016.

[12] A. R. S. Bahai, M. Singh, A. J. Goldsmith, and B. R. Saltzberg, "A new approach for evaluating clipping distortion in multicarrier systems," *IEEE J. Sel. Areas Commun.*, vol. 20, no. 5, pp. 1037–1046, Jun. 2002.

[13] Y. Chen, D. He, H. Wang, W. Yuan, and T. Q. S. Quek, "Joint channel and clipping amplitude estimation and signal detection for clipped OTFS," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 335–350, 2026.

[14] Q. Lu, L. Gui, and X.-Z. Fang, "A new scheme to mitigate the OFDM high PAR problem by minimizing the signal's nonlinear distortion caused by HPA," *IEEE Trans. Broadcast.*, vol. 52, no. 4, pp. 576–578, Dec. 2006.

[15] B. Benammar, N. Thomas, C. Poulliat, M.-L. Boucheret, and M. Dervin, "On linear MMSE based turbo-equalization of nonlinear Volterra channels," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, May 2013, pp. 4703–4707.

[16] Q. Shi, N. Wu, D. N. Nguyen, X. Huang, H. Wang, and L. Hanzo, "Low-complexity iterative detection for dual-mode index modulation in dispersive nonlinear satellite channels," *IEEE Trans. Commun.*, vol. 70, no. 2, pp. 1261–1275, Feb. 2022.

[17] Y. Zhang, B. Li, N. Wu, Y. Ma, W. Yuan, and L. Hanzo, "Message passing-aided joint data detection and estimation of nonlinear satellite channels," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 1763–1774, Feb. 2023.

[18] F. Luo and J. Wang, "PAPR reduction technology for ST-OCDM based joint radar-communication signals," in *Proc. IEEE MTT-S Int. Microw. Biomed. Conf. (IMBioC)*, Suzhou, China, May 2022, pp. 177–179.

[19] X. Lv, J. Wang, Z. Jiang, and W. Jiao, "A novel PAPR reduction method for OCDM-based radar-communication signal," in *Proc. IEEE MTT-S Int. Microw. Workshop Series 5G* Hardw. Syst. Technol. (IMWS-5G), Aug. 2018, pp. 1–3.

[20] M. ElHassan, M. Crussière, J.-F. Hélard, Y. Nasser, and O. Bazzi, "A novel statistical model of OFDM signals with optimal tone reservation PAPR reduction for EVM theoretical analysis," in *Proc. IEEE 29th Annu. Int. Symp. Pers., Indoor Mobile Radio Commun. (PIMRC)*, Sep. 2018, pp. 1–6.

[21] C.-L. Wang and Y. Ouyang, "Low-complexity selected mapping schemes for peak-to-average power ratio reduction in OFDM systems," *IEEE Trans. Signal Process.*, vol. 53, no. 12, pp. 4652–4660, Dec. 2005.

[22] W. Wang, T. Yu, S. Li, and B. Bai, "A simple phase rotation based PAPR reduction method for multicarrier Faster-than-Nyquist signaling," in *Proc. IEEE 98th Veh. Technol. Conf. (VTC-Fall)*, Oct. 2023, pp. 1–6.

[23] S. V. Zhidkov and R. Dinis, "Belief propagation receivers for near-optimal detection of nonlinearly distorted OFDM signals," in *Proc. IEEE 89th Veh. Technol. Conf. (VTC-Spring)*, Apr. 2019, pp. 1–6.

[24] C. Yang, X. Liu, Y. L. Guan, and R. Liu, "Fast GAMP algorithm for nonlinearly distorted OFDM signals," *IEEE Commun. Lett.*, vol. 25, no. 5, pp. 1682–1686, May 2021.

[25] T. Minka and J. Winn, "Gates," in *Proc. Adv. Neural Inf. Process. Syst.* Vancouver, BC, Canada: MIT Press, Dec. 2008, pp. 1073–1080.

[26] Y. Ma, N. Wu, W. Yuan, D. W. K. Ng, and L. Hanzo, "Joint channel estimation and equalization for index-modulated spectrally efficient frequency division multiplexing systems," *IEEE Trans. Commun.*, vol. 68, no. 10, pp. 6230–6244, Oct. 2020.

[27] T. Adali, P. J. Schreier, and L. L. Scharf, "Complex-valued signal processing: The proper way to deal with impropriety," *IEEE Trans. Signal Process.*, vol. 59, no. 11, pp. 5101–5125, Nov. 2011.

[28] Y. Ma, N. Wu, J. A. Zhang, B. Li, and L. Hanzo, "Parametric bilinear iterative generalized approximate message passing reception of FTN multi-carrier signaling," *IEEE Trans. Commun.*, vol. 69, no. 12, pp. 8443–8458, Dec. 2021.

[29] S. M. Kay, *Fundamentals of Statistical Signal Processing*. Upper Saddle River, NJ, USA: Prentice-Hall, 2017.

[30] H. Ochiai and H. Imai, "On the distribution of the peak-to-average power ratio in OFDM signals," *IEEE Trans. Commun.*, vol. 49, no. 2, pp. 282–289, Feb. 2001.

[31] X. Wang, X. Shen, F. Hua, and Z. Jiang, "On low-complexity MMSE channel estimation for OCDM systems," *IEEE Wireless Commun. Lett.*, vol. 10, no. 8, pp. 1697–1701, Aug. 2021.

[32] T. Zhan, T. Ma, Y. Yu, X. Lei, and J. Zhou, "An iterative channel estimation algorithm based on OMP for OCDM systems," *IEEE Commun. Lett.*, vol. 28, no. 7, pp. 1634–1638, Jul. 2024.

[33] S. Rangan, P. Schniter, and A. K. Fletcher, "Vector approximate message passing," *IEEE Trans. Inf. Theory*, vol. 65, no. 10, pp. 6664–6684, Oct. 2019.

[34] Y. Ma, N. Wu, K. Wu, and J. A. Zhang, "VAMP-based iterative equalization for index-modulated multicarrier FTN signaling," *IEEE Trans. Commun.*, vol. 71, no. 4, pp. 2304–2316, Apr. 2023.

[35] S. Rangan, "Generalized approximate message passing for estimation with random linear mixing," in *Proc. IEEE Int. Symp. Inf. Theory*, Jul. 2011, pp. 2168–2172.

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[36] Y. Ma, N. Wu, J. A. Zhang, B. Li, and L. Hanzo, "Generalized approximate message passing equalization for multi-carrier Faster-than-Nyquist signaling," *IEEE Trans. Veh. Technol.*, vol. 71, no. 3, pp. 3309–3314, Mar. 2022.
[37] L. Liu, S. Huang, and B. M. Kurkoski, "Memory AMP," *IEEE Trans. Inf. Theory*, vol. 68, no. 12, pp. 8015–8039, Dec. 2022.
[38] LTE Evolved Universal Terrestrial Radio Access (E-UTRA); *Base Station (BS) Radio Transmission and Reception*, Standard 3GPP TS 36.104 version 14.3.0 Release 14, ETSI TS, Apr. 2017.



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